What Happens in Paris, Does Not Stay in Paris: Trade Fairs and Search and Matching Frictions*

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Abstract

Search and matching frictions keep firms from forming international trade links. Trade fairs are a common, often subsidized remedy, yet causal evidence is scarce. We study Hungarian firms at the 1900 Paris World Exhibition. A trial exhibition in Budapest four years earlier had sorted firms into award classes that revealed their export potential before selection. This allows us to bound the effect with control firms from the award class above and below. We build a panel of 3,614 manufacturing firms for 1896–1909 from over 12,000 records across eleven historical sources. Participation raised the probability of exporting outside Austria-Hungary by 4.2 to 18.3 percentage points, against a 15 percent baseline, and the effect was still visible in 1909. The gains went to firms already exporting and were largest where there were many buyers and few domestic rivals. Patenting also rose, though innovation came with exporting rather than alongside it, and employment grew by 17.7 to 27.1 percent over eight years. These are private returns to selected participation: the fair is a congestible platform, so sending more similar firms need not deliver the same gain.

Keywords: buyer-supplier links, export promotion, trade fairs, search frictions, industrial policy, economic history, partial identification

JEL Codes: F13, F14, N84, O14, L25

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1 Introduction

Access to foreign markets allows firms to grow. How successfully firms reach those markets depends partly on their own productivity and managerial capability, but also on how easily they can find partners abroad. Exporting and importing require firms to search for foreign buyers and suppliers, verify that a prospective match is a good one, and finally persuade the partner to transact. Each step is costly and slow, so firm-to-firm links are hindered by substantial search and matching frictions (Allen, 2014; Eaton et al., 2021; Bergquist et al., 2024). Industry practitioners report that identifying a single suitable supplier can take several months (McKinsey & Company, 2021). Understanding the nature and magnitude of these frictions, and how they can be alleviated, is central to our understanding of international trade, the organization of global production networks, and the propagation of shocks across borders.

If these frictions are large, institutions that raise firm visibility and build trust between prospective partners can have first-order effects on trade. We contribute to this literature by studying the most common mechanism firms use to form new trade links: trade fairs. A trade fair is a platform that connects the two sides of a market: it gathers sellers and potential buyers in one place for a limited time, lowering the cost of meeting partners, inspecting products, and building trust. Trade fairs are pervasive: about 32,000 exhibitions took place worldwide in 2024/2025, attracting 4.7 million exhibiting companies (UFI, 2026). Participation is often subsidized by export promotion agencies (e.g., Broocks and Van Biesebroeck 2017; Volpe Martincus and Carballo 2010), yet rigorous causal evidence on their effectiveness remains scarce.

As a policy instrument, this platform has two distinctive features. First, it is selective. In contrast to transport and communication infrastructure improvements that broadly reduce the effective distance between all buyers and sellers (e.g., Jensen 2007; Jensen and Miller 2018; Allen 2014; Bernard et al. 2019), a trade fair limits participation to a chosen subset of firms, creating a more focused environment for link formation. Second, it is congestible. Unlike export promotion interventions that give individual firms targeted access to new buyers or suppliers (e.g., Atkin et al. 2017; Cai et al. 2024; Buus et al. 2025a; Munch and Schaur 2018; Carballo and Volpe Martincus 2008), trade fairs bring together large numbers of firms simultaneously, exposing participants to potential partners as well as their competitors. The benefit to an individual seller may therefore decline as similar sellers crowd the same product space and the pool of potential buyers thickens.

We study a unique historical experiment that allows us to confront the selection problem directly: the Hungarian government's selection process of firms for the 1900 Paris World

Exhibition. In 1900, Hungary was part of the Austro-Hungarian Monarchy (AHM), a polity with a common foreign policy and customs union but devolved fiscal, economic, and industrial policies. The setting offers three advantages. First, the government staged the 1896 Budapest Millennial Exhibition partly as a trial run, at which an independent jury awarded prizes based on criteria that explicitly included export potential. This provides us with an objective, ex ante, coarse measure of how well a firm would be expected to perform abroad, recorded three years before the Paris selection committee met. Second, that committee could not simply send the best firms; limited pavilion space in Paris and a mandate to represent industries and regions in balance meant that every award class contains both firms that were sent and firms that were not. Third, with travel costly and correspondence slow, firms had few alternatives to meeting in person, allowing us to isolate the effect of the fair from the private search channels that proliferate in modern settings.

Our research design exploits the first two features, a coarse ranking of export potential and imperfect sorting on it. If we observed export potential directly, we could condition on it and recover the treatment effect, regardless of how the committee chose. We observe only a binned version, and that coarseness is what turns a point estimate into an interval. We use the award classes to build two control groups for the same treated firms. The lower bound compares participants in a given award class to non-participants from the class immediately above, assigning treated firms controls with higher export potential and stacking the deck against finding an effect. The upper bound compares the same participants to non-participants from the class immediately below. Under a single assumption, that expected outcomes absent treatment are ordered across adjacent award classes, the two comparisons deliver causal bounds on the average treatment effect on the treated.

We report three estimates throughout the paper. The lower and the upper bound come from the two adjacent-class comparisons. The third is the within-class comparison, the estimate a researcher conditioning on the award as a proxy for quality would obtain. It identifies the effect only if selection within a class is unrelated to export potential, which is substantially more demanding than the ordering assumption the bounds rest on; hence, we report it as a third number rather than as our estimate. This is also what the design buys against the alternatives. Partial identification bounds that do not use an ex ante ranking are too wide to exclude zero here, and matching on observables reproduces the within-class comparison instead of correcting it.

Both parts of the design are testable, and we test them. First, if award classes rank export potential, pre-treatment characteristics should deteriorate across classes within both the treated and untreated groups alike; rank correlation tests bear this out. Second, the within-class estimate should lie between the two bounds, and it does. The balancing checks

then add one more thing: treated firms were positively selected within their own award class; hence, the within estimate is upward biased, and the true effect is likely to sit closer to the lower bound than to the upper bound. For this reason, wherever a mechanism is easier to show with a single number, we use the lower bound.

We build a panel of Hungarian firms from eleven independent historical sources: exhibition catalogs, government surveys, commercial directories, official gazettes, and patent databases. We digitize and link over 12,000 firm records using automated text extraction, geographic standardization, and multi-criteria entity resolution. The result covers 3,614 manufacturing firms between 1896 and 1909, with extra-AHM trade status, extra-regional trade status, employment, and patenting, though not every variable in every year.

Participation raised the probability of exporting outside the Austro-Hungarian customs union by 4.2 to 18.3 percentage points, against a baseline of 15 percent, and of selling outside the firm's own region by 7.6 to 20.6 points, against a baseline of 25 percent. Even the conservative lower bounds are increases of 28 and 30 percent relative to baseline, and the export effect was still visible in 1909. The gains went to firms that were already exporting, so the fair deepened internationalization rather than starting it.

Participation can work through several channels: by reducing search and matching frictions, by certifying quality and raising visibility, and by exposing firms to foreign products and technologies. We investigate these potential mechanisms behind the average effect and find that benefits were largest where the buyer side was thick and domestic rivals few. As we discuss in the conclusions, the heterogeneity matters greatly for policy design, especially as we only measure the private return to selected participation.

Our work contributes to four strands of literature. First, we contribute to the literature evaluating export promotion programs (e.g., Munch and Schaur 2018; Martincus and Carballo 2010; Carballo and Volpe Martincus 2008; Broocks and Van Biesebroeck 2017; Buus et al. 2025b; Van Biesebroeck et al. 2015; Hiller 2014; Görg et al. 2008; Schminke and Van Biesebroeck 2013; Lederman et al. 2010, 2016). While early studies often relied on difference-in-differences matching based on observable firm characteristics, more recent work exploits quasi-random variation in the selection processes of export promotion agencies or randomized access to support. The experimental evidence is sobering on durability: matching grants in Tunisia raised exports in the short run, but the effects faded within three years (Cadot et al., 2015), while a recent experiment on market-access subsidies finds more persistent gains (Ali et al., 2025). Randomized interventions, however, necessarily study modest numbers of program applicants in narrow settings and over short horizons. Our setting complements them on exactly these margins: we observe a near-census of one country's export-capable manufacturing sector spanning all industries and follow outcomes for up to

a decade after the intervention.

We extend this literature in three ways. First, we introduce a novel estimation approach that establishes "causal bands" for treatment effects, improving upon prior matching estimators by distinguishing between comparisons "close from above" and "close from below." Second, whereas export promotion agencies typically subsidize trade fair participation as part of a broader package of services, we evaluate the trade fair as a standalone intervention. Third, our setting allows us to examine not only the average effect on treated firms but also the role of the composition of participants, showing how exposure to competitors generates congestion effects while a thicker buyer pool amplifies gains.

Second, we relate to a growing literature studying firm-to-firm relationships. The empirical literature has focused on showing that exogenous reductions in search and matching frictions cause more matches. Part of the literature has used quasi-experiments that reduce transportation and communication cost to show evidence of search and matching frictions (e.g., Jensen 2007; Jensen and Miller 2018; Allen 2014; Bernard et al. 2019). Others employed RCTs to measure the effect of giving firms access to new suppliers (Cai et al., 2024) or buyers (Atkin et al., 2017; Bergquist et al., 2024). Face-to-face interaction plays a central role in this process: business travel predicts trade in relationship-intensive goods (Cristea, 2011), and in-person contact resolves not only search but also contracting frictions (Startz, n.d.), while online platforms that substitute for physical meetings sharply reduce the role of distance (Lendle et al., 2016). A complementary theoretical literature incorporates search and matching frictions or fixed costs of link formation to explain patterns of supplier-buyer relationships (e.g., Bernard et al. 2019; Eaton et al. 2022; Arkolakis et al. 2025; Miyauchi 2024); Lenoir et al. (2023) estimate that such frictions impede trade as much as traditional iceberg costs. We contribute to this literature by studying trade fairs – the most common mechanism firms use to form new links – which, unlike targeted interventions, simultaneously expose firms to potential partners and competitors, and unlike infrastructure improvements, operate in a selective, focused environment for matching. Our findings on the heterogeneous effects of trade fairs across product categories with different levels of competition and buyer concentration provide causal evidence on how search and matching frictions operate when firms are brought together in a concentrated environment.

Third, our work contributes to the growing body of quantitative research on how industrial policy influenced development during the first era of globalization (Juhász and Steinwender, 2024). Quantitative evaluations exist for temporary protection, infrastructure, intellectual property, colonial institutions, and tariffs (e.g., Juhász 2018; Hanlon 2020; Steinwender 2018; Alfaro et al. 2022; Romero 2023), but the role of trade fairs in shaping firm outcomes remains unexplored; prior research has instead used trade fair exhibitor data

as an alternative measure of innovation (Moser, 2005, 2012). We study a large-scale, centrally planned effort that transformed Hungary from a minor participant in 1878 into the fifth-largest national exhibitor at the 1900 fair.

Finally, our estimation approach builds on the literature using bounds to address selection problems in program evaluation, deriving credible treatment effect bounds under weaker assumptions (e.g., Manski 2003; Lee 2009; Manski and Pepper 2000; Tamer 2010; Ho and Rosen 2017; Blundell et al. 2007). Our setting shares key features with this literature: we face a selection problem where exhibiting firms likely differ from non-exhibitors in unobservable ways. Like Manski and Pepper (2000), we exploit an ex ante ranking variable – the Budapest exhibition awards – that partially predicts treatment assignment. However, our approach innovates in how we construct bounds. Rather than using trimming procedures as in Lee (2009), or comparing across levels of the monotone treatment selection pattern as in Manski and Pepper (2000), we exploit the discrete nature of our award categories to construct two distinct control groups (same vs. lower-tier awards). This approach offers several advantages: it provides clear economic intuition by comparing treated firms to those of similar versus superior export potential and incorporates two built-in validation tests.

The paper is organized as follows. Section 2 provides the historical context. Section 3 describes the panel of Hungarian firms we build by digitizing and linking eleven historical sources. Section 4 sets out the empirical strategy, deriving the lower and the upper bound and the within-class estimate, and stating what each requires. Section 5 takes the design to the setting and reports the validation: industrial representation among participants, balancing tests, and the rank correlations implied by the ordering assumption. Section 6 presents the export effects. Section 7 turns to mechanisms: for whom the fair worked, competition and market thickness, certification through the Paris awards, and innovation. Section 8 follows the consequences for employment and firm survival, the persistence of the export effect to 1909, and what remains visible today. Section 9 concludes with the implications for export promotion policy.

2 Hungarian firms in the 1900 Paris World Exhibition

The 1900 Paris *Exposition Universelle* was the largest world's fair to date, celebrating the achievements of the 19th century and showcasing visions for the 20th. Running from April to November 1900, it attracted over 48 million visitors, akin to France's entire population. The exhibition featured 76,000 exhibitors from 56 countries displaying innovations ranging from moving sidewalks (*trottoirs roulants*) and early motion pictures to diesel engines and passenger trolleybuses. Exhibitors were organized into 18 broad categories based on the

general theme of the products they exhibited, and further subdivided into 121 subcategories corresponding to narrower product groups.

The exhibition took place across multiple halls in different parts of Paris, many of them near the *Champs de Mars*. Firms in the same broad category shared a hall; for instance, one hall was dedicated to "Mines et métallurgie" (mining and metallurgy). Within each hall, exhibits were arranged by country, with Hungary typically placed alongside Austria and Belgium.

For participating countries, the exhibition served purposes well beyond cultural diplomacy and national prestige. Governments explicitly recognized its function as, in effect, a trade fair: the vast concentration of international visitors, including merchants, industrialists, and potential customers, created unprecedented opportunities for establishing business connections (Tamilia, 2007). Firms showcased their products and innovations, demonstrated product capabilities, negotiated terms with prospective buyers, and used the occasion for networking through meetings, lunches, dinners, and organized events. Contemporary accounts describe the exhibition grounds as bustling with commercial negotiations alongside the display of innovations.

The exhibition's commercial importance was amplified by the communication constraints of the era. International business relationships typically required either costly travel or slow correspondence by mail. The telegraph, while revolutionary for time-sensitive information, was most effective for trade in homogeneous products that could be codified, but limited in facilitating trade in goods that required physical inspection (Juhász and Steinwender, 2018). The exhibition thus offered a rare opportunity for face-to-face meetings that could build trust and enable detailed product demonstrations.

Between 1867 and 1918, Hungary was part of the Austro-Hungarian Monarchy (AHM), a dual state in which the Austrian Emperor also served as King of Hungary (see a map in Appendix E). The empire maintained a common foreign policy and customs union, but Hungary had its own parliament, finances, and economic and industrial policies.

The AHM was a late industrializer compared to Western European powers such as Great Britain, France, or Germany, resulting in substantially lower per capita incomes between 1870 and 1913. However, within the monarchy, Hungary grew faster than Austria, achieving an annual GDP per capita growth rate of 1.32%, closer to France (1.45%) than Austria (1.05%) (Schulze, 2000). This reflected Hungary's industrialization, which had begun in the late 19th century: the employment share in manufacturing, crafts, and utilities rose from 6.6% in 1869/1870 to 11.1% in 1910, though it still lagged behind Austria (21.4%) and Germany (29.4%) (Schulze, 2007).

The Hungarian government perceived the Paris Exhibition as an ideal opportunity to

promote Hungarian exports (Kun, 2019), especially since it marked the first time Hungary participated in a world fair as an exhibitor independent of Austria. The Hungarian Royal Minister of Commerce, Baron Ernő Dániel, noted that "*[...] the 1900 Paris Exhibition [...] will be a meeting place for foreigners flocking there from all parts of the world, and thus a favorable opportunity for establishing commercial connections. The Hungarian Royal Commissioner, who [...] is [...] organizing the representation of the countries of the Hungarian Crown [...] will support the establishment of such commercial connections*" (*Hungarian Industrial Journal*, 1897, ed. 2).

To maximize the economic benefits of the exhibition, the Hungarian government prepared extensively and strategically. Preparations began as early as 1887, thirteen years before the Paris Exhibition, and included establishing institutions and procedures for selecting firms. A key element was the use of the 1896 Budapest Millennial Exhibition, held four years before Paris, as a trial exhibition to identify the country's most competitive products and firms (Székely, 2015).

To ensure broad participation in the trial exhibition, the Budapest exhibition was widely advertised in newspapers and posters, supplemented by outreach through industrial associations and chambers of commerce. The effort paid off: with 9,650 exhibitors from the Hungarian Kingdom proper participating, the exhibition provided near-comprehensive coverage of Hungary's export-relevant manufacturing sector.

At the trial exhibition, an independent jury assessed each exhibit's export potential and awarded four prize categories in descending order: gold, silver, bronze, and honorable mention.[1] The jury was organized into 34 product-group panels with a total of 480 members, half elected by exhibitors and half appointed, following formal regulations.[2] An exhibit received an award only upon a majority vote of the relevant panel.[3] Jury members and their firms were excluded from receiving prizes.[4]

After the trial exhibition, a separate selection committee, distinct from the 1896 award jury, was tasked with choosing firms for Paris. While the primary goal was to promote Hungarian exports by selecting firms with high export potential, internal documents show that the committee also sought balanced representation across industries and regions, aiming to present a broad cross-section of Hungarian production. In addition, exhibition space in Paris was limited and transport costs were high, so not all promising candidates could be accommodated. Within some product categories, space constraints meant that even firms

[1]For simplicity, we relabeled the names; the original names and descriptions of the awards are found in Appendix F.1.2.

[2]For details on jury selection, see Appendix F.1.1.

[3]For more details on criteria and justification for awards, see Appendix F.1.3 and F.1.4.

[4]Note that for this reason, we exclude jury members from our sample of firms.

with demonstrated export potential were excluded, while others were included to maintain sectoral balance.

Overall, Hungary sent approximately 3,000 exhibitors across all sectors to Paris, making it the fifth-largest national exhibitor at the fair. Beyond granting participation, the government covered all costs for selected exhibitors, including transportation of exhibits to Paris.

Contemporary accounts make the commercial function of the fair explicit. Mór Jókai, then Hungary's most widely read author, was in Paris during the exhibition and filed dispatches for the Budapest press. Alongside the tourism and the hotel prices, he reported the commercial outcome in the plainest possible terms: Kölber's light carriage, he wrote, was ordered twelve times, and orders for Zsolnay majolica came in a hundredfold.[5] This is the object our export outcomes measure four years later, observed at the moment of formation: a small number of firms leaving the fair with a foreign order book.

3 Data

A drawback of using a historical event is the lack of a ready-made dataset. In fact, we created and combined eleven datasets, listed in Table 1, to construct a panel dataset of Hungarian firms. All data were digitized by us from historical archival sources.

The most complex technical challenge was entity resolution across the independently collected datasets. Firm names vary in spelling, abbreviation, and format across sources, requiring a combination of matching algorithms and extensive manual checking of potential matches. We created correspondence tables across dataset pairs and consolidated them into a master identification system enabling both balanced and unbalanced panel specifications. Further details on the digitization process and entity resolution methods are provided in Appendix Section G.

In the following, we describe how we construct the dataset used in our baseline analysis combining the different sources.

Sample. The *Budapest Exhibition Catalog* documented 9,650 exhibitors from Hungary at the 1896 Budapest Millennial Exhibition (excluding Croatia-Slavonia and Bosnia-Herzegovina) across 23 exhibition groups: one historical group and 22 contemporary groups broadly corresponding to economic sectors. 58 detailed product categories are nested within the 22 contemporary exhibition groups.

The distribution of exhibitors across groups is shown in Table D.2. We considered only commercially oriented manufacturing firms that entered the competition (3,736) and oper-

[5] Mór Jókai, *Pesti Hírlap*, June 1900.

Table 1: Overview of historical data sources

Short name (translated)	**Original name**	**Year**	**Key variables**
Budapest Exhibition Catalog	Az 1896-iki Ezredéves Országos Kiállítás általános katalógusa	1896	Product group, firm characteristics, incl. extra-AHM trade status and employment
Budapest Exhibition Brochure	1896-iki ezredéves országos kiállítás közleményei	1896	Awards from Budapest exhibition
Central Gazette of Hungarian Firms	Központi Értesítő	1896–1904	Firm exits
Paris Exhibition Catalog	Catalogue Général Officiel	1900	Treatment, product group
Hungarian Museum of Commerce Survey 1	Kiviteli czímtár	1898	Extra-regional trade status and employment
Hungarian Museum of Commerce Survey 2	Kiviteli címtár	1904	Extra-regional trade status and employment
Hungarian Museum of Commerce Survey 3	Beszerzési Források	1909	Extra-regional trade status
Austro-Hungarian Export Compass	Export-Compass, commerzielles Jahrbuch für d. Interessenten des österreichisch-ungarischen Ausfuhr-Handels	1905	Extra-AHM trade status
Imperial Privileges	Privilegiensammlung des österreichischen Patentamts	up to 1899	Patents
Hungarian Patent Office	Szellemi Tulajdon Nemzeti Hivatala	up to 1905	Patents
PATSTAT	EPO PATSTAT Austria	up to 1905	Patents
Google Patents	Google Patents for Austria, United States, Germany, France, Great Britain and Switzerland	up to 1905	Patents

ated throughout (see details in Section G). We find that 122 firms exited during this period, bringing our baseline sample to 3,614 firms.[6]

Baseline characteristics. The *Budapest Exhibition Catalog* provides exhibitor-level information, including names, addresses, exhibited products, and exhibition group, for nearly all participants. For a subset of exhibitors, the catalog also records firm-level characteristics such as workforce size. Where firm characteristics are missing from the *Budapest Exhibition Catalog*, we supplement them with the *Hungarian Museum of Commerce Survey 1* from 1898 where available.

Unobserved export potential: award category. We match these firms to the official jury results published in the *Budapest Exhibition Brochure* at the exhibition's close to obtain information on the awards granted to each firm. There were four award categories: gold, silver, bronze, and honorable mention.[7]

Treatment: participation in the Paris Exhibition. We identify treated firms in our baseline sample as Paris exhibitors if we successfully match it to an entity listed in the *Paris Exhibition Catalog*.

Extra-AHM trade. This dummy measures whether a firm appears in the *Austro-Hungarian Export Compass* from 1905, compiled by the Austrian Ministry of Commerce in partnership with regional chambers of commerce. This directory lists all firms that exported outside the Austro-Hungarian customs union in the reference year, organized by product category. Notice that this is a demanding measure of exporting, as it requires exporting outside the empire; we do not have a measure of exporting outside the Hungarian Kingdom.

Extra-regional trade. This indicator variable measures whether a firm appears in the *Hungarian Museum of Commerce Survey 1* covering 1898 (pre-treatment), in the *Hungarian Museum of Commerce Survey 2* covering 1904 (post-treatment), or in the *Hungarian Museum of Commerce Survey 3* covering 1909 (post-treatment).[8] The stated objective of the survey was to compile a registry of firms that were either exporting outside the Austro-Hungarian customs union, exporting outside the borders of the Hungarian Kingdom but within the Austro-Hungarian customs union (e.g., from Budapest to Slavonia, now part of Croatia, or

[6] In Section 8.2, we examine the effect of the Paris exhibition on firm exits and survival.

[7] For details, see Appendix F.1.2.

[8] These surveys were conducted in 1894, 1897-98, 1904, 1909 and 1914. The earliest survey from 1894 only covered a limited sample of firms. We mainly focus on the directory appearance in 1904, which is the closest subsequent year to the Paris trade fair, and the 1909 version is used as a long-term outcome.

to Pozsony, now Bratislava, Slovakia, or from Kassa now Kosice in Slovakia to Fiume now Rijeka in Croatia), or selling outside of their region inside the Hungarian Kingdom.[9]

Employment. The *Hungarian Museum of Commerce Survey 2* provides data on employment levels for 1904 (post-treatment), albeit for only a subset of firms.

Patenting. We combined four patent databases to construct a comprehensive database of patenting by Hungarian firms during our sample period: *Imperial Privileges*, the *Hungarian Patent Office*, *PATSTAT Austria*, and *Google Patents* (covering six countries). From these, we constructed indicator variables for whether a firm held at least one patent in each of three periods: before 1895, 1896-1899, and 1900-1905.

Key variables summary. Summary statistics on all variables are presented in Table 2, separately for baseline characteristics and variables measured post-treatment. Of the Hungarian firms in our sample, 15% participated in the 1900 Paris Exhibition. At baseline, 15% of firms were already exporting in 1896, and 25% reported being capable of exporting in 1898. Detailed firm characteristics such as workforce size, installed horsepower, and output value are available for about 13% of firms. Among firms with available employment data, the range spans from 1 to over 17,000 workers, with a mean of 172. Seven percent of firms operate more than one establishment in Hungary, and 4% are organized as joint-stock companies. The average firm age is 26 years, and 9% of firm have patented by 1895.[10]

Turning to the award categories from the Budapest Exhibition, gold was the most selective, received by only 3.7% of firms. Another 11.2% received silver, followed by a larger bronze category comprising 27.9% of firms. Nearly a third received an honorable mention, and 26.1% received no award.

Paris exhibitors and product categories Data on non-Hungarian exhibitors comes from 20 volumes of the official product-group catalogs, organized into 18 groups and 121 Paris product category. We then use LLMs to summarize each Paris product category's jury overview and identify exhibitor types and representative products, producing structured Paris product category-level summaries that serve as inputs for the buyer-score and input-output analysis. For details on data parsing and variable creation, see Section G.9.1 in the Appendix.

[9] For additional details, see Appendix Section G.

[10] Appendix Table D.1 presents baseline characteristics separately for Paris exhibitors and non-Paris exhibitors.

Table 2: Summary statistics

Variable	N	Mean	SD	Min	Max
Treatment					
Paris exhibitor dummy	3,614	0.15	0.35	0	1
Baseline characteristics (1896, unless indicated otherwise)					
Extra-AHM trade status	3,614	0.15	0.36	0	1
Extra-regional trade status (1898)	3,614	0.25	0.44	0	1
Number of workers	1,193	172.40	783.47	1	17,165
Share of skilled workers	992	0.76	0.39	0.006	8.67
Installed horsepower	652	224.89	997.51	1	15,700
Output (in HUF)	541	688,042	1.842 mil	400	14 mil
Multi-location firm	3,614	0.07	0.25	0	1
Joint-stock company	3,614	0.04	0.19	0	1
Firm age	1,391	25.95	30.76	1	510
Patent holder (1895)	3,614	0.09	0.28	0	1
Patenting indicator (1896-1899)	3,614	0.11	0.31	0	1
Gold medal indicator	3,614	0.037	0.190	0	1
Silver medal indicator	3,614	0.112	0.315	0	1
Bronze medal indicator	3,614	0.279	0.448	0	1
Honorable mention indicator	3,614	0.310	0.463	0	1
No award indicator	3,614	0.261	0.440	0	1
Post-treatment outcomes (1904, unless indicated otherwise)					
Extra-AHM trade status (1905)	3,614	0.11	0.31	0	1
Extra-regional trade status	3,614	0.24	0.43	0	1
Number of workers	796	192.01	740.57	2	17,000
Patenting indicator (1900-1905)	3,614	0.12	0.32	0	1

4 Empirical strategy

The central challenge in estimating the causal effect of trade fair participation is selection: firms that exhibit are not drawn at random, either because they choose to attend or because they are selected to do so. Our setting is unusual in that it provides a coarse, ex ante measure of export potential, recorded by an independent award jury at a separate exhibition three years before the Paris selection committee met. We exploit this measure to bound the causal effect of participation at the trade fair.

4.1 Causal bounds

To fix ideas, let $Y_i(1)$ and $Y_i(0)$ denote firm i's export outcome with and without participation after the fair, let D_i indicate participation, and let $\tau_i = Y_i(1) - Y_i(0)$ be the firm-specific treatment effect, so that the observed outcome is $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$. Our object of interest is the average treatment effect on the treated (ATT), $E[\tau_i \mid D_i = 1]$. A comparison of mean outcomes between participants and non-participants does not identify it, because it also contains a selection term:

$$\underbrace{E[Y_i \mid D_i = 1] - E[Y_i \mid D_i = 0]}_{\text{naive difference}} = \underbrace{E[\tau_i \mid D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_i(0) \mid D_i = 1] - E[Y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

In the literature, the selection effect is typically positive. However, in our setting, the sign of the selection term is not obvious ex ante, because the Paris selection committee looked for export potential but also had to consider other constraints.

Let $\theta_i := E[Y_i(0) \mid i]$ denote firm i's *export potential*: how well the firm would be expected to do in foreign markets absent participation. If we observed θ_i, we could condition on it and recover the treatment effect, no matter how the Paris selection committee chose, since by construction participants and non-participants with the same θ have the same expected counterfactual outcome. Selection would no longer be a problem, as long as both participants and non-participants are observed at each value of θ.

A common problem in this literature is that researchers do not observe θ_i, and neither do we. What is unusual in our setting is that we observe a coarsened version A_i of it. Formally, let $A_i \in \mathcal{A} = \{1, \ldots, 5\}$ denote the binned export potential of firms, where $A_i = 1$ denotes the highest export potential. In our application, the role of A_i is played by four prize levels awarded by the Budapest award jury based on criteria that explicitly included export capability, plus the residual category of firms that received no award from the preceding trial exhibition; we therefore refer to A_i as award classes throughout. We defer the discussion of why awards are a valid proxy for export potential to Section 5 and first establish the

assumptions that A_i must satisfy for the estimation of our causal bounds.

The key idea in our research design is to use the award classes to construct two distinct control groups for treated firms, yielding an upper and a lower bound on the ATT. The lower bound compares treated firms to untreated firms from the award category immediately above, "stacking the deck" against finding an effect by assigning treated firms a control group with higher export potential and thereby providing a conservative estimate. The upper bound compares treated firms to untreated firms from the award category immediately below. Because these control firms have a lower export potential, this comparison is likely upward biased.

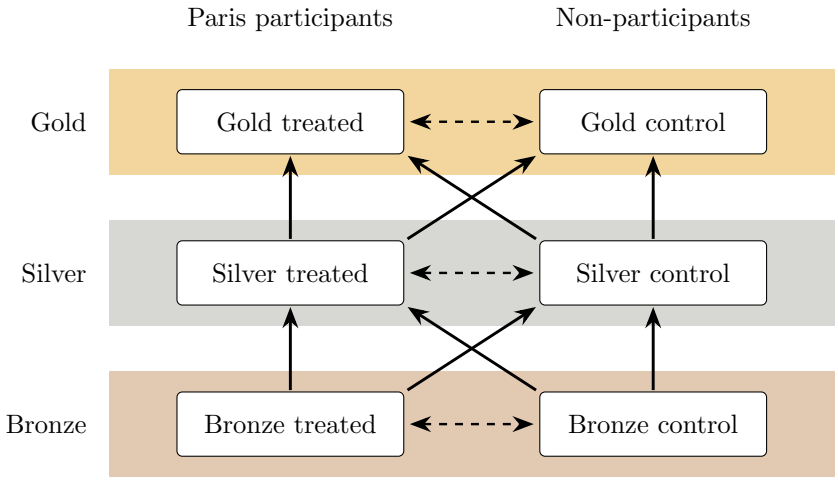


Figure 1: Using award classes to construct causal bounds: illustration

Notes: Solid arrows point to higher export potential. Double-pointed dashed arrows indicate an arbitrary relationship with respect to export potential.

Figure 1 illustrates this idea. The award classes are shown as rows and provide a ranking: firms in better award classes have higher export potential than firms in a lower award class.[11] In each award class, some firms were selected to participate at the Paris exhibition, while others were not. The ordering with respect to export potential established by award classes implies that both treated firms in Gold as well as control firms in Gold have higher export potential than either treated or untreated firms in Silver and lower award classes. This is illustrated by solid arrows pointing in the direction of higher export potential. For the lower bound, we compare the treated firms in silver to the control firms in gold, which have higher export potential. For the upper bound, we compare the treated firms in silver to the control

[11]The strongest assumption on award classes would require *all* firms in better award classes to have higher export potential than *all* firms in lower award classes. For our bounds, we only need a weaker version of this ordering, as we will discuss below.

firms in bronze, which have lower export potential.

Notice that to make these two comparisons, it is not necessary to take any stance on whether treated firms have higher, lower or the same export potential as untreated firms *within* an award class. This is illustrated by dashed, double-pointed horizontal arrows. Selection within an award class could be positive, negative, or absent, and it may even differ across award classes. Furthermore, while intuitive in our setting, it is not necessary to assume that *all* firms in a higher award class have higher export potential than *all* firms in a lower award class, as the comparisons in Figure 1 are between the mean of the treated firms in an award class and the mean of the control firms in the class above or below, not between individual firms. Formally, we require:

Assumption 1 (Mean-ordered award classes)**.** *For any* $d, d' \in \{0, 1\}$*,*

$$E[Y(0) \mid D = d, A = a - 1] \;\geq\; E[Y(0) \mid D = d', A = a] \quad \forall a \in \mathcal{A} \setminus \{1\}.$$

Let $\tau^a := E[Y(1) \mid D = 1, A = a] - E[Y(0) \mid D = 1, A = a]$ denote the award-class specific ATT. Comparing Paris participants from award class a to non-participants from the better award class $a - 1$,

$$\beta^a_{lower} \;:=\; E[Y(1) \mid D = 1, A = a] \;-\; E[Y(0) \mid D = 0, A = a - 1],$$

and to non-participants from the worse award class $a + 1$,

$$\beta^a_{upper} \;:=\; E[Y(1) \mid D = 1, A = a] \;-\; E[Y(0) \mid D = 0, A = a + 1],$$

yields causal bounds on the ATT:

Lemma 1 (Causal bounds)**.** *Under Assumption 1,*

$$\beta^a_{lower} \;\leq\; \tau^a \;\leq\; \beta^a_{upper},$$

where β^a_{lower} *is defined for* $a \in \{2, \ldots, 5\}$ *and* β^a_{upper} *for* $a \in \{1, \ldots, 4\}$*.*

Proof in Appendix A.[12] In our research design, the coarsened bins are enough to eliminate selection as the fundamental difficulty in our setting. Neither bound requires knowing how the Paris selection committee chose: it may have had private information about export

[12]Note that it uses only part of Assumption 1, the cross-treatment group and cross-award group comparisons indicated by the diagonal arrows in Figure 1. We state Assumption 1 for all subgroup means as it is a natural generalization, and because the ordering of the remaining pairs, indicated by the vertical solid arrows in Figure 1, delivers additional validation tests in Section 4.4.

potential, responded to lobbying, or applied industry and regional quotas. None of this violates Assumption 1, because none of it moves a firm from one award class to another.[13]

Instead of selection, coarseness is the fundamental challenge in our research design: the bins deliver an interval of treatment effects rather than a point estimate of the treatment effect, which observing θ_i would deliver. Had the Budapest award jury graded firms more finely, adjacent classes would differ less in export potential and the identified set would shrink, until in the limit of continuous grading the design reduces to matching on export potential itself. That limit is not attainable, and not only because we lack the finer award classes: had the Paris selection committee chosen strictly by export potential, sufficiently fine award classes would contain either only participants or only non-participants, but not both. What generates the overlap between participants and non-participants within award classes is precisely that the committee did not select on export potential alone, but was restricted. Coarseness and overlap pull in opposite directions, and our design exploits the fact that both are present.

4.2 Within-award estimate

Researchers might be inclined to condition on award class as a proxy for export potential. This comparison identifies the ATT if the selection of firms to participate in Paris *within* award classes was orthogonal to their export potential:

Assumption 2 (No selection within award classes)**.** *Within award classes, selected firms have the same export potential as non-selected firms:*

$$E[Y_i(0) \mid D_i = 1, A_i = a] \; = E[Y_i(0) \mid D_i = 0, A_i = a], \quad \forall a.$$

In Figure 1, this assumption implies that the dashed, double-pointed horizontal arrows represent equal export potential. Comparing Paris participants to non-Paris participants from the same award class a,

$$\beta^a_{within} \; := \; E[Y(1) \mid D = 1, A = a] \; - \; E[Y(0) \mid D = 0, A = a],$$

provides a point estimate of the ATT:

Lemma 2 (Within-award class comparison)**.** *Under Assumption 2,*

$$\beta^a_{within} = \tau^a \quad \forall a.$$

[13] See also Appendix Figure C.1 for an additional illustration of how Lemma 1 accommodates positive, negative or neutral selection within award classes.

Proof in Appendix A. Assumption 2 holds if the constraints faced by the Paris selection committee were unrelated to export potential. The industrial and regional representativeness and the space constraints suggested by the historical narrative have the potential to support this. It fails if the Paris selection committee chose the firms with the highest export potential *within* an award class, or if firms with lower export potential *within* an award class lobbied successfully to be sent. But it does not fail if selection or lobbying differentially favors sending firms *across* award classes, as they are controlled for. Unlike Assumption 1, which rests on what the Budapest award jury was asked to do, Assumption 2 is a restriction on the Paris selection committee, taking awards as given.

Assumptions 1 and 2 together are more demanding than Assumption 1 alone, but they deliver more: the bounds continue to hold, and the point estimate they contain is the treatment effect. If instead selection within classes is positive, β^a_{within} overstates τ^a, and the bounds are what remains. We therefore report the within estimate alongside the bounds throughout.

4.3 Implementation

We estimate each comparison by OLS with fixed effects for the relevant comparison group,

$$\Delta Y_i \;=\; \beta\, D_i \;+\; \sum_g \delta_g \mathbf{1}\{G_i = g\} \;+\; \gamma\, Y_{i,pre} \;+\; FE_p \;+\; \epsilon_i, \tag{1}$$

where the definition of the comparison groups G_i determines which set of estimands β aggregates: (i) the award classes themselves, for β_{within}; (ii) groups consisting of treated firms from award class a and untreated firms from class $a-1$, for β_{lower}; and (iii) groups consisting of treated firms from class a and untreated firms from class $a+1$, for β_{upper}. ΔY_i denotes the change in the outcome before versus after the Paris trade fair. We control for the baseline level of the outcome, $Y_{i,pre}$, in case changes in the outcome differ by the initial level, and product category fixed effects FE_p. Product category fixed effects are the 58 most detailed product categories of the 1896 Budapest catalog, which nest within the 22 contemporary exhibition groups; standard errors are clustered at the same level.

The three comparisons are defined on different sets of treated firms. Treated firms in the top award class have no better class to draw control firms from and drop out of the lower bound; treated firms without an award have no worse class to draw control firms from and drop out of the upper bound. To ensure that differences across the three estimates are not driven by the change in sample, we also report all three on the common sample of treated firms in award classes 2 to 4, together with the control firms each comparison assigns to

them. The bounds are then bounds on the ATT among these firms rather than among all participants. Whether the excluded top-awarded firms gained more or less is not obvious: they may have gained more, because higher quality attracts more buyers, or less, if firms that already had foreign contacts faced diminishing returns.[14] Our estimates are informative about the marginal supported firm, which is the margin a government expanding an export promotion program would move.

4.4 Validation tests

The credibility of the causal bounds rests on Assumption 1, and the credibility of the within-estimate rests on Assumption 2, neither of which is directly testable. This parallels other research designs: in difference-in-differences, for instance, researchers routinely conduct balancing tests on pre-treatment variables to support the plausibility of their identifying assumptions. These types of balancing checks can also be implemented in our research design and correspond to testing for differences in pre-treatment characteristics between treated firms in lower to untreated firms in higher award classes for the lower bound[15], between treated firms in higher to untreated firms in lower award classes for the upper bound[16], and between treated and untreated firms in the same award class for the within-estimate[17].

However, our research design allows us to implement two additional validation tests. Ex ante, we test whether pre-treatment characteristics follow the ordering of award classes separately for treated and untreated firms:[18]

Lemma 3 (Rank test)**.** *Under Assumption 1, for any* $d \in \{0, 1\}$*,*

$$E[Y(0) \mid D = d, A = a - 1] \;\geq\; E[Y(0) \mid D = d, A = a] \quad \forall a \in \mathcal{A} \setminus \{1\}.$$

We test both orderings on observable pre-treatment characteristics using Spearman rank correlation tests.

Ex post, we test the ordering of the estimated treatment effects. The three comparisons differ only in the award class supplying the control firms, so Assumption 1 implies that the

[14] Table 5 speaks to this indirectly: estimating effects separately by prior export status, even the lower bounds are larger for previous exporters than for previous non-exporters.

[15] Testing the relationship given by the diagonal arrows pointing to upper right in Figure 1 and corresponding to the part of Assumption 1 needed in Lemma 1 for the lower bound.

[16] Testing the relationship given by the diagonal arrows pointing to upper left in Figure 1 and corresponding to the part of Assumption 1 needed in Lemma 1 for the upper bound.

[17] Testing the relationship given by the horizontal dashed arrows in Figure 1 and corresponding to Assumption 2.

[18] These two tests correspond to the two additional inequalities stated in Assumption 1 that are not used in Lemma 1 and correspond to the vertical solid lines in Figure 1.

within estimand is sandwiched between the lower and the upper bound:

Lemma 4 (Sandwich test)**.** *Under Assumption 1,*

$$\beta^a_{lower} \;\leq\; \beta^a_{within} \;\leq\; \beta^a_{upper}.$$

Failure of ordering would reject Assumption 1. The condition is necessary but not sufficient: the inequalities holding is consistent with, but does not guarantee, the validity of the assumption.

All proofs in Appendix A.

4.5 Relationship to alternative estimation approaches

Our design adds a novel identifying assumption to the partial identification framework of Manski (2003). We share the idea of exploiting a covariate that ranks firms by their potential outcomes, as in the monotone instrumental variable (MIV) of Manski and Pepper (2000), but suggest an alternative identification assumption: Assumption 1 requires the monotone ordering of expected potential outcomes with respect to the covariate to hold for all specific subgroups the causal bounds compare, while the MIV restriction requires it to hold on average for all firms. Our alternative assumption delivers three advantages. It yields tighter bounds, which we illustrate in Appendix B.5. It generates testable implications, the rank test and the sandwich test of Section 4.4. And it sidesteps the search for control firms that closely resemble the treated: rather than seeking exact matches, we construct control groups that bound the effect from above and below, which may be easier to defend.[19]

A second alternative research design is to match on the firm characteristics we observe. We implement this in Appendix **??** and show the results together with the Manski bounds results for comparison in Appendix B.1, but our results reveal that these matching estimands are still upward biased in our setting.

5 Application to the historical setting

The distribution of Paris participants across award classes in Table 3 is consistent with the two features that our research design requires. The first is that award classes carry information about export potential: participation declines monotonically in award rank, from 61.5% among gold medalists to 6.8% among firms without an award. The second is

[19]Appendix B gives the formal comparison of our identifying assumptions to bounded outcomes, MTR, MTS, and MIV.

overlap: participation is strictly between 0 and 100% in every award class. Only 14.7% of firms were sent to Paris, and even in the highest award class fewer than two thirds went. Selection tracked the awards without being fully determined by them, which leaves every award class with both participants and non-participants.

Table 3: Distribution of firms across 1896 award classes and participation in the Paris exhibition

	Firms in sample		**Paris exhibitors**		**Non-Paris exhibitors**	
Award category	**Number**	**Share**	**Number**	**Share**	**Number**	**Share**
Gold medal	135	3.7%	83	61.5%	52	38.5%
Silver medal	405	11.2%	147	36.3%	258	63.7%
Bronze medal	1,007	27.9%	161	16.0%	846	84.0%
Honorable mention	1,122	31.0%	78	7.0%	1,044	93.0%
No award	945	26.1%	64	6.8%	881	93.2%
Total	3,614	100%	533	14.7%	3,081	85.3%

Three features of the 1896 Budapest exhibition support Assumption 1. First, export potential was an explicit criterion in the jury's mandate.[20] Second, the awards were determined by an independent expert jury, separate from the committee that later selected firms for Paris. Third, the awards were assigned in 1896, three years before any Paris participant was selected.

Two threats could violate the ordering required by Assumption 1. The first is government influence over the 1896 Budapest awards. If politically connected firms received better awards than warranted by their export potential and were also more likely to be sent to Paris, the treated firms in a given award class would have lower export potential than their class implies. This violates the identification assumptions on the upper bound, which requires treated firms in class a to be no worse than untreated firms in class $a+1$. The lower bound is, if anything, helped: it already assumes treated firms in class a to be no better than untreated firms in class $a-1$, and if the true export potential of the former is even worse, this assumption is even more likely to be satisfied. Notice that only influence over the awards is a threat here. Influence over the Paris selection does not violate Assumption 1, and it is in fact one of the reasons award classes contain both participants and non-participants. The second threat is grading error by the Budapest award jury. Random grading error does not affect the ordering of subgroup means. Assumption 1 is only violated if the grading error is correlated with Paris selection, that is, if the firms the jury underrated within a class were also the ones the committee later chose.

In either case, our research design can accommodate both some correlated grading error

[20] For details, see Appendix F.

and some political influence. Assumption 1 does not require all firms in a higher award class to have higher export potential than all firms in a lower one, only that the subgroup means remain ordered, so incorrectly graded or politically favored firms are a problem only if there are enough of them to reverse that ordering.

5.1 Industrial representation among Paris participants

Assumption 2 concerns the 1900 Paris selection committee rather than the Budapest award jury. It holds if the committee's selection within award classes was unrelated to export potential. Its constraints, a balanced representation of industries and regions, together with limited exhibition space and the high transport costs of exhibits, have the potential to break the link between selection and export potential within award classes.

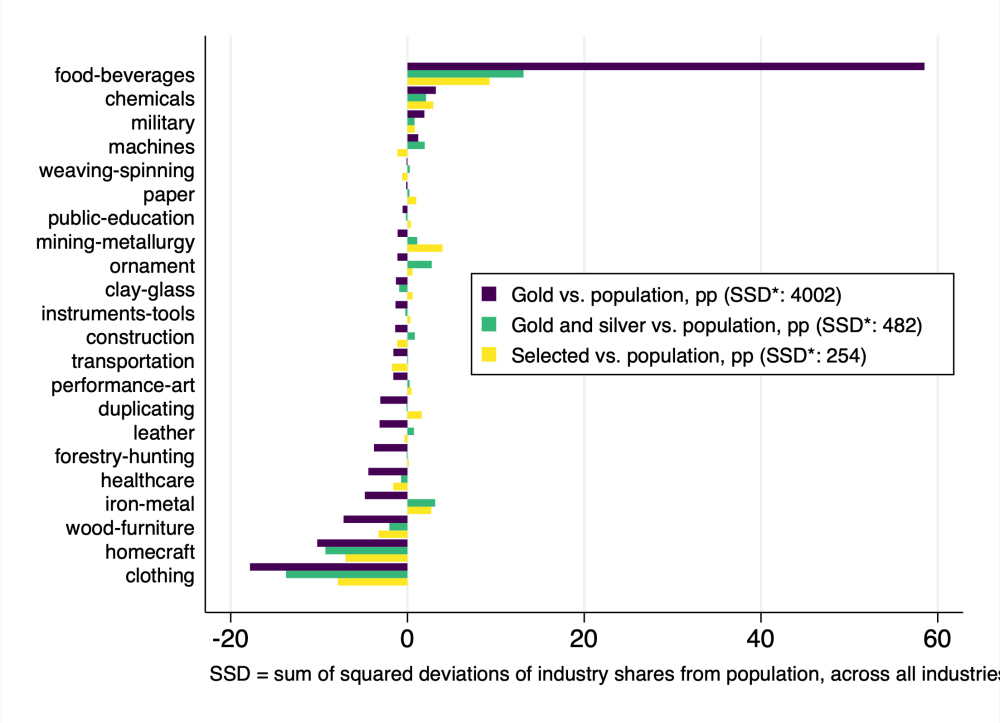


Figure 2: Deviation of industry shares from the population distribution: gold medalists, gold and silver medalists, and actual Paris participants

To verify that industrial representation was a binding constraint for the Paris selection committee, consider a thought experiment. How representative would the group sent to Paris have been had the committee selected purely on export potential? We construct two such counterfactual groups: the 135 gold medalists alone, and the 540 gold and silver medalists, a number close to the 533 firms actually sent. Figure 2 plots, for each industry, the difference between its share within a group and its share in the population of Budapest exhibitors.

Both counterfactual groups show substantial deviations. Food and beverages are strongly over-represented, consistent with Hungary's early success in flour milling, while clothing is under-represented, consistent with the concentration of textile and clothing production in the Austrian and Bohemian parts of the Habsburg empire, which left Hungarian manufacturers at a competitive disadvantage (Ránki, 1964). The firms the committee actually sent deviate far less.[21] We obtain a similar pattern for regional representation.[22] The Paris selection committee therefore traded export potential against industrial and regional balance, which is one reason every award class contains both participants and non-participants.

5.2 Balancing tests

For β_{lower} to be a valid lower bound, treated firms in an award class must have equal or lower export potential than control firms in the class above. For β_{upper} to be a valid upper bound, they must have equal or higher export potential than control firms in the class below. For β_{within} to identify the treatment effect, treated and control firms in the same award class must have the same export potential.

[21] Summarizing the deviations by the sum of squared distances from the population distribution, the value falls from 4,002 for gold medalists and 482 for gold and silver medalists to 254 for Paris participants.

[22] At the county level, the sum of squared distances falls from 43 for gold medalists and 19 for gold and silver medalists to 6 for Paris participants.

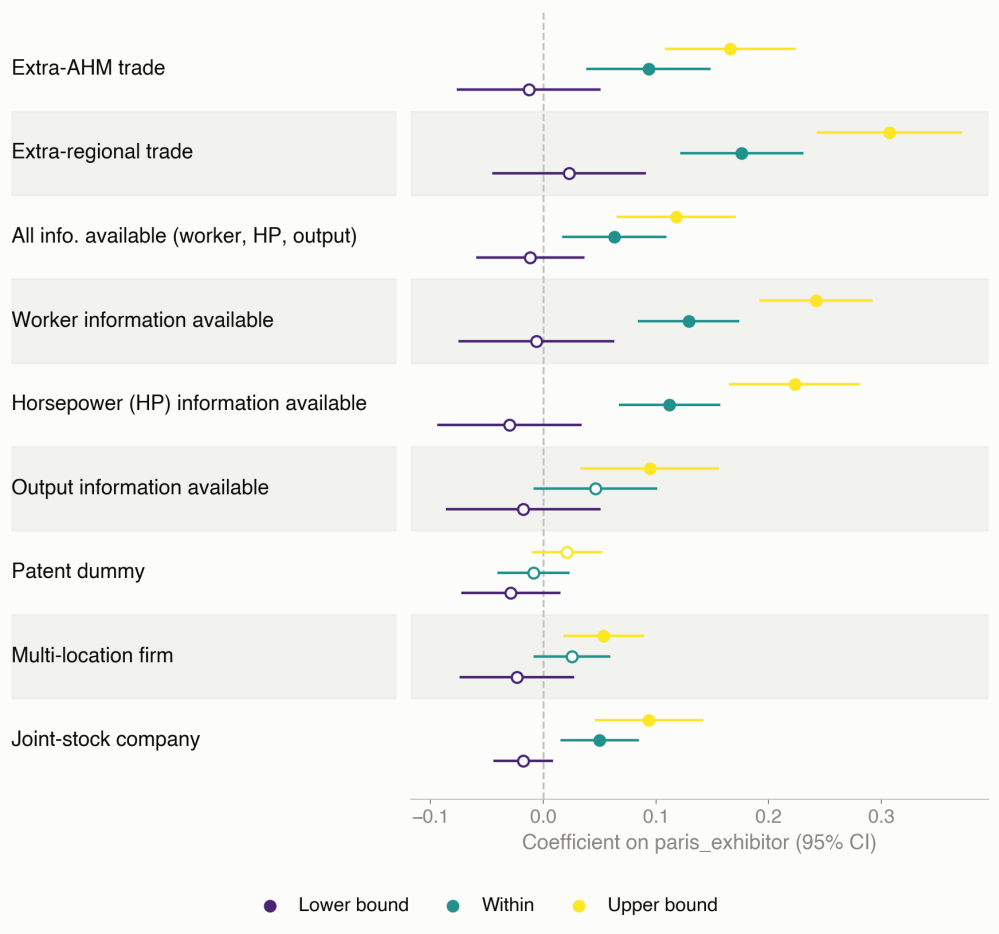


Figure 3: Balancing tests on pre-treatment indicator variables for lower bound, within-estimate, and upper bound

Notes: The figure plots the coefficient on Paris participation from estimating equation (1) with pre-treatment indicator variables as dependent variables. Each coefficient is the mean difference between treated firms in award class a and untreated firms in award class $a-1$ (lower bound), a (within), or $a+1$ (upper bound).

We test whether these relationships hold for pre-treatment characteristics in Figure 3. The green upper-bound estimates show that treated firms look better than their controls on most characteristics, with almost all differences statistically significant. The blue lower-bound estimates show the opposite: treated firms look similar to or slightly worse than their controls, and none of the differences is significantly positive. Both patterns are what Assumption 1 requires.

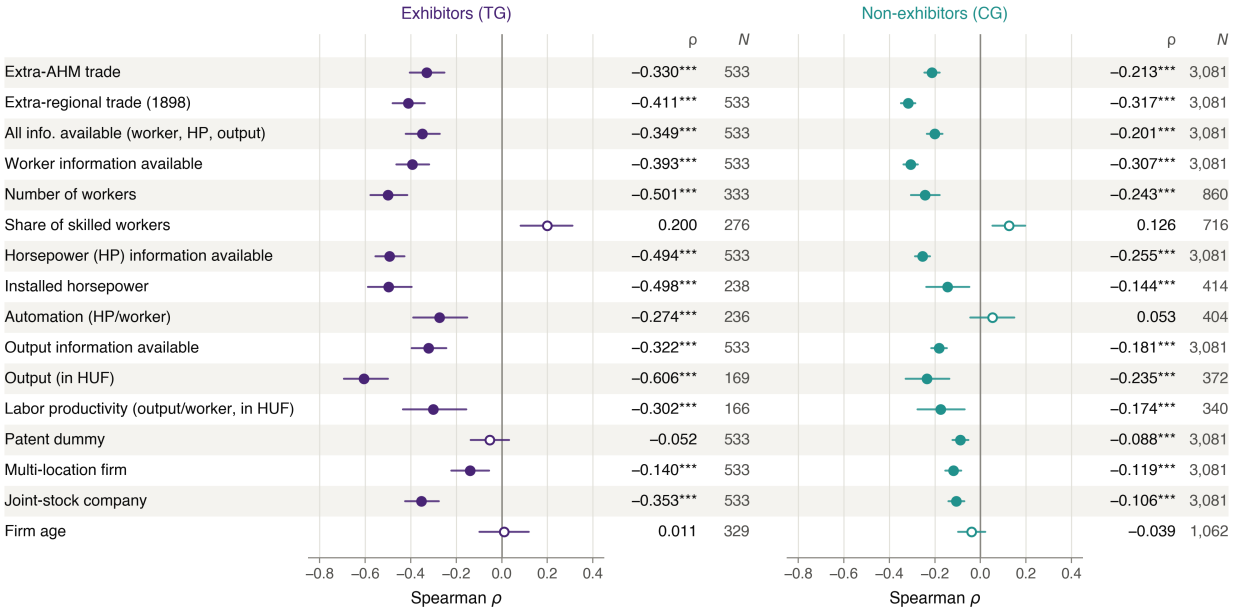
The evidence is less supportive of Assumption 2. Treated firms within award classes are significantly better on several characteristics, indicating that selection within classes was not orthogonal to export potential and that any residual sorting was positive. The within estimate is therefore best read as an upward-biased estimate rather than a point identified

causal effect. This is precisely the situation our research design is built for: it recovers bounds where conditioning on award class alone does not identify the causal effect. Moreover, every within-class difference lies between the corresponding lower and upper bound difference, the same ordering Lemma 4 predicts for the estimated treatment effects.

Results for continuous variables, which are available only for subsets of firms, show the analogous pattern and are reported in Appendix Figure C.2.

5.3 Rank test

Figure 4: Spearman rank correlation by treatment status between pre-treatment firm characteristics and award classes



Notes: The figure plots Spearman rank correlation coefficients between pre-treatment firm characteristics and award classes a, separately for Paris exhibitors and non-Paris exhibitors, as stated in Lemma 3. The null hypothesis is $H_0 : \rho \geq 0$; the one-sided alternative is $H_1 : \rho < 0$. Stars denote one-sided p-values for this directional test: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Horizontal lines denote 95% confidence intervals constructed via Fisher z-transformation.

Figure 4 reports the rank test, which checks whether pre-treatment characteristics become worse in lower award classes, separately for the treatment and control group. Consistent with Lemma 3, almost all characteristics are significantly negatively correlated with award class for both the treatment and control group. Firms in better award classes were larger, more mechanized, more productive, and more likely to export. Skill share is the only exception

and could indicate lower-ranked firms being less mechanized and more reliant on craft work, but this variable is observed for only a quarter of firms.[23]

6 The causal effect of the Paris exhibition on exporting

Table 4 reports our estimated effects of participating in the Paris exhibition on exporting, using either the change in extra-AHM trade or in extra-regional trade as the dependent variable in regression (1). Columns (1) and (3) show that Paris participation increased extra-AHM trade by between 4.2 and 18.3 percentage points. Relative to the baseline extra-AHM export share of 15% (Table 2), the lower bound implies an increase of 28%, while the upper bound implies more than a doubling. By construction, the lower bound is conservative, since it assigns treated firms a control group from a better award class and stacks the deck against finding a positive treatment effect. Yet the lower bound is positive and statistically different from zero: even the most conservative estimate thus provides evidence of a positive causal effect of trade fair participation. The within estimate in column (2) is 10.1 percentage points. However, since the balancing tests in Section 5.2 suggest that treated firms within award classes were positively selected, the within estimate is most likely upward biased.

Our estimation strategy passes the ex post validation test specified in Lemma 4: the upper bound estimate exceeds the lower bound estimate, and the within estimate lies inside the bounds. The last row reports formal (one-sided) tests of this ordering; both differences from the within estimate are positive and statistically significant, as Assumption 1 requires.

Columns (4) to (6) report a positive treatment effect of Paris on extra-regional trade, bounded by 7.6 and 20.6 percentage points, or 30% and 82% relative to the baseline extra-regional trade share of 25% (Table 2). The ordering of the three estimates again holds.[24]

[23] Appendix Figure C.3 shows a negative correlation between the skill share and both mechanization and firm size in the sample where both are observed.

[24] Appendix Table D.4 replicates Table 4 on the common sample.

Table 4: The effect of the Paris exhibition on exporting

	Extra-AHM trade, 1905			Extra-regional trade, 1904		
	(1) Lower bound	(2) Within	(3) Upper bound	(4) Lower bound	(5) Within	(6) Upper bound
Paris exhibitor	0.042*	0.101***	0.183***	0.076***	0.133***	0.206***
	(0.024)	(0.022)	(0.029)	(0.025)	(0.023)	(0.027)
Pre-treatment outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.478	0.473	0.475	0.313	0.311	0.314
Difference from within estimate	0.059***		0.082***	0.057***		0.073***
	(0.015)		(0.017)	(0.014)		(0.012)

Notes: Each column estimates equation (1). Column (1) and (4) include all firms from the Budapest exhibition; column (2) and (5) restrict the sample to firms used in the lower-bound estimation; column (3) and (6) restrict the sample to firms used in the upper-bound estimation. The pre-treatment outcome is extra-AHM trade in 1896 in columns (1) to (3) and extra-regional trade in 1898 in columns (4) to (6). The regressions calculating the difference from the *within* estimate are described in Appendix D.3. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Overall, we find strong evidence that the Paris trade fair increased exporting. In Appendix Figure B.1 we compare our causal bounds to alternative estimators. Our bounds are substantially tighter than the three commonly used Manski bounds and exclude zero for both outcomes, which makes them more informative for policy makers who must decide whether to use trade fairs as an export promotion tool. The matching estimators deliver estimates close to or above the within estimate, but the balancing checks in Section 5.2 indicate that the within estimate is upward biased. Matching on observables therefore reproduces the within-class comparison rather than correcting for its bias.

7 Mechanisms: for whom and through which friction

Having shown that participation raised exporting, we now ask how. We start with *for whom* the fair worked, separating firms that were already exporting from those that were not.

We then look at three channels. First, *search and matching*: being in proximity to many potential buyers lowers the cost of finding a counterparty. Second, *certification and visibility*: getting an award increases visibility at the Paris trade fair and beyond. Third, *learning and exposure*: companies spent up to seven months in Paris and had time to look for exhibited products and processes, they could see innovation by all exhibitors and learn from it.

7.1 For whom: new vs. established exporters

An important policy question is who gains from a trade fair. We separate firms that were already exporting in 1896 from those that were not and ask whether the Paris trade fair worked differently for the two groups.

Trade fairs may affect these groups differently. For non-exporters, trade fairs reduce the cost of finding an initial foreign buyer, but converting a match into a transaction may require existing capabilities, such as an exportable product or the managerial capacity to fulfill foreign orders. For established exporters, trade fairs may provide additional partners that help sustain export activity during adverse conditions or enable expansion into new markets. Meeting partners may also cement or extend existing relationships, thus helping to ensure the survival of exporting status.

The estimates presented so far pool two potentially different margins: firms entering export markets for the first time and firms that were already exporting before the exhibition. To separate these effects, we estimate the Paris effect separately for prior exporters (β_1) and non-exporters (β_2):

$$\Delta Y_i = \beta_1 \, D_i \times Y_{i,pre} + \beta_2 \, D_i \times (1 - Y_{i,pre}) + FE_b + Y_{i,pre} + FE_p + \epsilon_i \tag{2}$$

Table 5 reports the results for extra-AHM trade (columns 1, 2, and 3) and extra-regional trade (columns 4, 5 and 6), with within estimates plus upper and lower bounds for each. Two results stand out. First, the effects are substantially larger for prior exporters: 15.8 vs 8.3 percentage points for extra-AHM trade and 17.4 vs 9.6 percentage points for extra-regional trade for the within estimation. Second, the lower bounds for non-exporters are only 2.4-2.9 percentage points and are statistically insignificant.

The export effects of Table 4 are thus driven by firms that were already exporting, for which the fair helped sustain or expand existing activity, rather than by non-exporters using the exhibition to start. Trade fairs, on this evidence, deepen internationalization rather than create it from scratch.

Table 5: Treatment Effect on Export Performance, by Previous Exporter Status

	Extra-AHM trade, 1905			Extra-regional trade, 1904		
	(1)	(2)	(3)	(4)	(5)	(6)
	Lower bound	Within	Upper bound	Lower bound	Within	Upper bound
Paris exhibitor × previous exporter	0.111*	0.158***	0.280***	0.142***	0.174***	0.236***
	(0.065)	(0.054)	(0.052)	(0.041)	(0.032)	(0.031)
Paris exhibitor × non-exporter	0.024	0.083***	0.149***	0.029	0.096***	0.175***
	(0.025)	(0.027)	(0.036)	(0.033)	(0.031)	(0.040)
Pre-treatment outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.282	0.314	0.312	0.433	0.471	0.474

Notes: Column (1) and (4) include all firms from the Budapest exhibition; column (2) and (5) restrict the sample to firms used in the lower-bound estimation; column (3) and (6) restrict the sample to firms used in the upper-bound estimation. The pre-treatment outcome is extra-AHM trade in 1896 in columns (1) to (3) and extra-regional trade in 1898 in columns (4) to (6). Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

7.2 Search and matching: competition and market thickness

Unlike interventions that give individual firms targeted access to new buyers or suppliers, a trade fair brings many firms together at once. It is therefore a congestible platform: expanding participation changes both the competition and the buyer thickness each firm faces. Two forces are at play. Firms may find potential buyers, but they also compete for attention. In this section, we estimate how the treatment effect varies along these two margins, measured by the number of competitors a firm faced at the fair and by market thickness, proxied by the number of potential buyers among other exhibitors. This tells us how the private return to a heterogeneous firm population would shift as the policy scales.

To investigate this mechanism, we need to leverage our information on all Paris exhibitors, as introduced in Section 3 (See G.11.1 in the Appendix for additional details). Now we start by defining competition and buyer potential.

The competition variable measures how many other exhibitors a company competed with within its Paris product category j, N_Hun_j for Hungarian firms and N_Int_j for others. This is multiplied by a within Paris product category buyer share π_j. Thus,

$$\text{Hungarian Competition}_j = (1 - \pi_j) \times N_Hun_j$$

International competition is calculated the same way.

We define cross-Paris product category *Buyer Potential* as a probability weighted sum of potential international buyers. In the main model, we excluded French firms since an increase in French imports from Hungary might result from learning about the French market outside the scope of the exhibition. We then use these probabilities to weight the number of potential buyers each Hungarian exhibitor faced. For a Hungarian firm exhibiting in Paris product category j, each Paris product category k to which j has a forward link contributes $P_{jk} \times N_k$ potential buyers, where N_k is the number of exhibitors in Paris product category k; summing over all such k gives $\sum_{k\,:\,j\to k} P_{jk} \times N_k$.

Exhibitors within the firm's own Paris product category j are not automatically treated as competitors: where the Paris product category summary identifies a share of same-Paris product category exhibitors as downstream buyers rather than rivals – for instance firms integrated across a later processing stage – these are added as well. Denoting by π_j the share of Paris product category-j exhibitors identified as likely buyers rather than competitors, and by N_j the total number of exhibitors in Paris product category j, the own-Paris product category contribution is $\pi_j \times N_j$. For a firm in Paris product category p, *buyer potential* is:

$$\text{buyer potential}_j = \sum_{k\,:\,j\to k} P_{jk} \times N_k + \pi_j \times N_j$$

Having defined competition as well as buyer potential as our measures of market thickness, we analyze how the treatment effect varies along these two dimensions by estimating:

$$\begin{aligned} \Delta Y_i \;=\; & \beta\, \mathrm{D}_i + \gamma \mathrm{D}_i \times \ln(\text{Hungarian competition})_i \\ & + \delta \mathrm{D}_i \times \ln(\text{international buyer potential})_i + FE_b + Y_{i,pre} + FE_p + \epsilon_i \end{aligned} \tag{3}$$

Both interactions are defined only for Paris exhibitors, so γ and δ measure how a participant's return varies with the competition and the buyer potential of its Paris product category.

Table 6 shows that both buyers and competitors affect how strongly participation at the Paris fair increases exporting: Hungarian firms that are part of product categories with many other Hungarian firms benefit less from being in Paris, possibly because this divides the attention of buyers and makes it harder for specific firms to stand out from the competition. Conversely, Hungarian firms in product categories with many potential buyers benefit more.

To illustrate, consider the marginal effects. First, hold the expected number of international buyers constant and compare firms at the bottom and top quartiles of the competitor distribution, where ln(Hungarian competitors) is 2.64 at p25 and 3.81 at p75. Moving across

Table 6: Effect of competition and potential buyers on change in extra-AHM trade, 1896-1905

	Extra-AHM trade, 1905					
	Lower bound	Within	Upper bound	Lower bound	Within	Upper bound
	(1)	(2)	(3)	(4)	(5)	(6)
Paris exhibitor	-0.053	0.034	0.038	0.036	0.082	0.192
	(0.099)	(0.081)	(0.093)	(0.114)	(0.102)	(0.122)
Paris * ln(international buyer potential)	0.034***	0.033***	0.039***	0.037***	0.035***	0.042***
	(0.009)	(0.009)	(0.010)	(0.010)	(0.009)	(0.010)
Paris * ln(Hungarian competition)	-0.047*	-0.056***	-0.048**	-0.032	-0.049**	-0.027
	(0.027)	(0.020)	(0.021)	(0.032)	(0.022)	(0.026)
Paris * ln(international competition)				-0.024	-0.012	-0.037*
				(0.022)	(0.018)	(0.022)
Pre-treatment outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.484	0.479	0.482	0.485	0.479	0.483

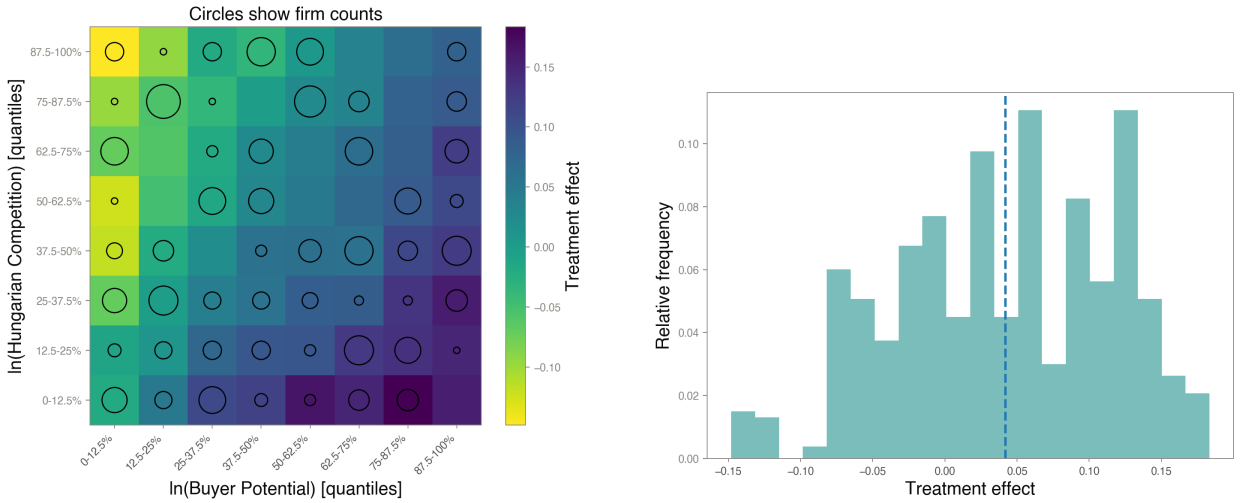
Notes: Columns (1) and (4) restrict the sample to firms used in the lower-bound estimation; column (2) and (5) include all firms from the Budapest exhibition; column (3) and (6) restrict the sample to firms used in the upper-bound estimation. The pre-treatment outcome is extra-AHM trade in 1896. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

this 1.17 log-unit range *lowers* the export probability by 5.5 percentage points (4.7×1.17). This represents over 130 percent of the average treatment effect (4.2 percentage points). Second, repeat the comparison for buyers, holding competition constant, where p25 is 5.90 and p75 is 7.27. The 1.36 log-unit difference corresponds to a 4.6 percentage point higher export probability.

Both margins are therefore large relative to the average effect, and their relative importance is stable across the three bound samples. The buyer channel strengthens as the sample widens, and the competition channel is strongest in the within sample, so the conclusion does not hinge on which bound sample is used. Figure 5 plots the implied treatment effect for the lower bound sample by quantiles of competitors and buyers.

Both figures show substantial variation. Figure 5a shows a small group of firms for which the exhibition had no effect, or a slightly negative point estimate, and an equally small group with very high effects, where few Hungarian competitors and many potential buyers coincide.

Overall, firms in product categories with fewer Hungarian rivals and more potential foreign buyers at the exhibition benefited most from Paris participation in terms of exporting. A small share of Paris exhibitors that faced many domestic competitors and few potential buyers did not benefit at all. This result is important as it indicates that search and matching frictions are important, and that competition and greater buyer potential could alter the cost-benefit analysis of being at a large event.



(a) Heatmap of treatment effects

(b) Histogram of treatment effects

Figure 5: Heterogeneous treatment effects by competition and international buyers (lower bound)

Note. Figure 5a shows the joint distribution of heterogeneous treatment effects based on column (1) of Table 6. Black hollow circles indicate the number of firms in each cell, with circle area proportional to cell frequency. Missing cells (with zero firms) are filled by iterative averaging of neighboring cells to obtain a smooth surface. Figure 5b shows a histogram of treatment effects, with the weighted mean indicated by a dashed line.

7.3 Paris awards: global reach via certification

Firms could win awards at the Paris exhibition, as they could at the Budapest trial exhibition. These awards were decided by international juries composed of French and foreign experts. In this section, we ask whether winning an award was a separate mechanism that raised exports, for example, by certifying quality, signaling brand value, or increasing visibility. A product carrying the insignia of a Paris award may have reached buyers who never attended the exhibition, extending the effect beyond the contacts made in Paris and potentially generating a more sustained response.

We test this mechanism by estimating the treatment effect separately for Paris participants who won an award and those who did not. Columns (1) to (3) of Table 7 show that award winners benefited substantially more than participants who did not win. Firms with any Paris award gained 7.4 percentage points in the lower bound, 13.7 points in the within-award comparison, and 20.0 points in the upper bound. Participants without an award gained much less, and nothing at all in the lower bound. Columns (4) to (6) split participants further by the rank of the Paris award. The estimated effect is monotonic in rank: in the lower bound, 22.1 percentage points for a Grand Prix or gold medal, 7.4 points for silver or bronze, and 0.3 points for an honorable mention or no award. Benefits are thus

concentrated among the high-visibility distinctions.

Table 7: Treatment effect of certification on extra-AHM trade

	Extra-AHM trade 1905					
	Lower	Within	Upper	Lower	Within	Upper
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: any vs. no award						
Paris × any Paris award	0.074**	0.137***	0.200***			
	(0.030)	(0.027)	(0.034)			
Paris × no Paris award	−0.000	0.044	0.155***			
	(0.032)	(0.030)	(0.039)			
Panel B: award quality						
Paris × high Paris award				0.221***	0.248***	0.324***
				(0.062)	(0.063)	(0.068)
Paris × medium Paris award				0.074*	0.136***	0.205***
				(0.037)	(0.031)	(0.035)
Paris × low/no Paris award				0.003	0.052**	0.137***
				(0.023)	(0.022)	(0.029)
Pre-treatment outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.480	0.475	0.476	0.483	0.477	0.479

Notes: Any Paris award is an indicator equal to one if a firm received any award at the Paris Exposition. *High Paris award* includes Grand Prix and gold medals; *Medium Paris award* includes silver and bronze medals; and *Low/No Paris award* includes honorable mentions and firms that received no award. Columns (1) and (4) restrict the sample to firms used in the lower-bound estimation; columns (2) and (5) include all firms from the Budapest exhibition; columns (3) and (6) restrict the sample to firms used in the upper-bound estimation. The pre-treatment outcome is extra-AHM trade in 1896. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

The stronger effects for award winners are consistent with a certification channel, but that channel can operate in two ways. An award may help a firm reach buyers who were never in Paris, if the distinction remained attached to the product after the fair closed. It may also improve matching at the fair itself, by making the firm more visible to the buyers already present.

To separate the two, we ask whether awards reinforce the role of market thickness, allowing the treatment effect to vary with the buyer potential a firm faces:

$$
\begin{aligned}
\Delta Y_i \;=\; & \beta\, \mathrm{D}_i \times \mathrm{NoAward}_i + \gamma\, \mathrm{D}_i \times \mathrm{Award}_i \\
& + \delta\, \mathrm{D}_i \times \mathrm{NoAward}_i \times \ln(\text{Buyer Potential})_i \\
& + \theta\, \mathrm{D}_i \times \mathrm{Award}_i \times \ln(\text{Buyer Potential})_i \\
& + FE_b + Y_{i,pre} + FE_p + \epsilon_i
\end{aligned}
\tag{4}
$$

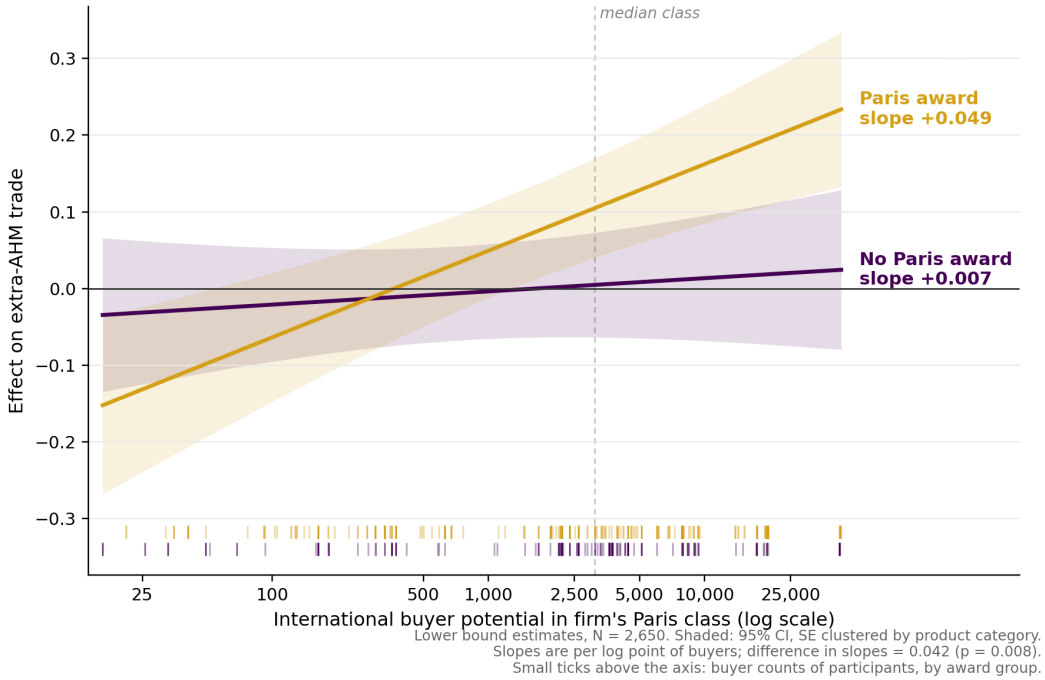


Figure 6: Awards as mediator for market thickness

Notes: Lower bound estimates, with 95% confidence interval. Based on estimates reported in Appendix Table D.7. Ticks mark where the mass of firms sits.

Figure 6 shows that Paris award winners benefit more where they face more potential buyers, while the effect for participants without an award is flat in buyer potential.[25] Rather than a separate certification channel, this points to certification working through search and matching. An award raised exports only where buyer potential was high, which is what one expects if its role was to help a firm stand out among the many sellers competing for the same buyers. It is the counterpart of the congestion result in Section 7.2: there, a firm's return fell as more Hungarian rivals competed for the same buyers; here, an award is what

[25]See Appendix Table D.7 for estimated coefficients and additional extensions. Parameter estimates are qualitatively stable across lower, within, and upper bound estimates. The statistical significance of the difference between slopes drops below 5% for some modeling choices.

set a firm apart from those rivals, and so it was worth most where there were more buyers to win over.

7.4 Innovation: learning or market size?

The third mechanism is innovation, as firms in Paris observe other companies' exhibitions and may benefit from new knowledge. While participation in Paris may raise innovation, we need to separate two possible channels because they carry different policy implications. Under *learning and exposure*, the fair is a place to see and possibly adapt or build upon foreign technology. This requires no export order, so innovation can rise on its own. Under *demand and scale*, the fair delivers customers, and the larger market raises the return to innovation enough to justify its fixed cost (Lileeva and Trefler, 2010; Bustos, 2011); here, innovation arrives coupled with exporting, not instead of it.

The two channels differ in two observable dimensions. First, *who* innovates: learning permits innovation without export expansion, while market size does not. Second, *what* firms patent: learning should pull the content of a firm's patents toward the technology on display, while market size should leave the direction unchanged.

We measure innovation through patenting, using the four patent databases described in Section 3 to construct an indicator of whether a firm applied for or registered at least one patent in the post-exhibition period of 1900–1905.

We start by comparing exhibitors and non-exhibitors in terms of patenting as the outcome. Columns (1) to (3) of Table 8 report our estimates: Paris exhibitors were 4.2–9.1 percentage points more likely to patent after the exhibition. Against a baseline patenting rate of 9.8% among Paris participants (Appendix Table D.1), this is a large increase[26]. As a placebo exercise, columns (4) to (6) show no pre-exhibition effect, which rules out both additional invention while preparing exhibits and defensive patenting to protect ideas from rivals at the fair.

[26]The increase comes from firms that held no patent in 1895 (Appendix Table D.5): the fair turned non-innovators into first-time patentees rather than making existing innovators more prolific.

Table 8: Treatment effects on patenting

	Patenting, 1900-1905			Patenting, 1896-1899 (as placebo)		
	Lower bound	Within	Upper bound	Lower bound	Within	Upper bound
	(1)	(2)	(3)	(4)	(5)	(6)
Paris exhibitor	0.042*	0.061***	0.091***	−0.011	0.010	0.035
	(0.023)	(0.017)	(0.019)	(0.024)	(0.019)	(0.021)
Patenting status 1895	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.216	0.209	0.207	0.214	0.205	0.198

Notes: The outcome is an indicator variable that equals 1 if the firm applied for or registered at least one patent in the respective time period. Columns (1) and (4) restrict the sample to firms used in the lower-bound estimation; column (2) and (5) include all firms from the Budapest exhibition; column (3) and (6) restrict the sample to firms used in the upper-bound estimation. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Having established an effect on patenting, we turn to the market-size channel. We decompose outcomes into four mutually exclusive categories: firms that both export and patent, firms that only export, firms that only patent, and firms that do neither. We concentrate on prior exporters, the group for which Section 7.1 found the export effect to be concentrated. Figure 7 reports lower bound estimates.

The two marginal effects at the top are those already reported: an 11.3 percentage point increase in extra-AHM trade and a 5.9 percentage point increase in patenting. Paris raises the probability of exporting without patenting by 4.3 percentage points and of doing both by 7.0 percentage points, and these two sum to the effect on exporting.

For non-exporters that patent was unaffected by Paris: with an estimate of −0.012 at the lower bound (and small at the within and upper bounds as well). Innovation, in other words, arrived together with exporting rather than on its own. This result is more in line with the market-size and learning-by-exporting channels than with technology exposure operating independently of the export margin. Appendix Table D.8 reports the full decomposition.

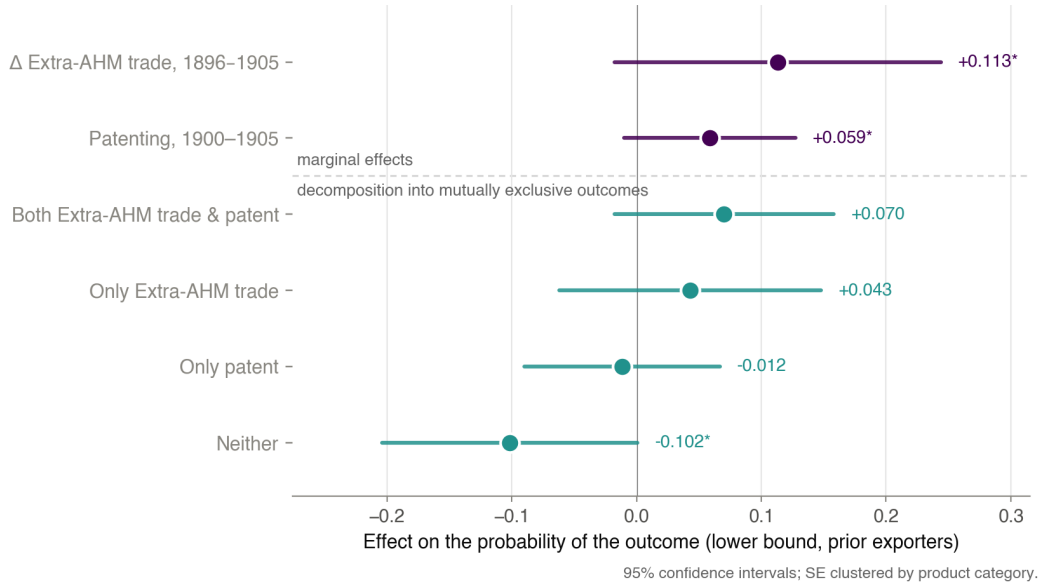


Figure 7: Joint effects on exporting and patenting, prior exporters

Notes: Lower bound estimates for firms that were already exporting in 1896, with 95% confidence intervals. The upper two estimates are the marginal effects on extra-AHM trade and on patenting. The lower four decompose them into mutually exclusive outcomes, defined on extra-AHM trade in 1905 and patenting in 1900–1905. All specifications control for baseline extra-AHM trade status, baseline patenting status and their interaction, and include product fixed effects and lower-bound group dummies. Standard errors are clustered by product.

The second test uses the content of the patents rather than their count. If Hungarian firms took technology home from Paris, the patents they filed afterward (1900 to 1905) should look more like the patents of the firms they might have encountered there. We collect the patent stock of all non-Hungarian firms exhibiting in Paris using the *Google Patents* records of 5 major countries (United States, Germany, France, Great Britain, and Switzerland) and measure the textual similarity between each Hungarian firm's patent title and that set, taking either the maximum or the mean across a Hungarian firm's patents.[27] Most firms in our sample never patent, so we assign non-patenting firms the mean similarity of non-exhibitors, which defines the outcome for the full sample.

We compare the mean and the maximum patent similarity metrics for Hungarian firms that exhibited to those that did not, controlling for pre-exhibition patenting status and patent similarity. Table 9 shows no effect. While Paris exhibitors increase patenting, we do not find any evidence of changing patent topics towards patents filed by firms in Paris in the five years following the exhibition. Firms did not seem to have altered their patenting

[27] We used embeddings of the patent titles to transform them into vectors and cosine similarity to calculate pairwise similarities.

strategy based on what they might have seen.

Table 9: Effect of trade fair attendance on patent similarity

	Patent similarity to Paris patents, 1900–1905					
	Mean similarity			Maximum similarity		
	(1)	(2)	(3)	(4)	(5)	(6)
Paris exhibitor	-0.000	-0.000	-0.001	-0.001	-0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Patent similarity, up to 1895	Yes	Yes	Yes	Yes	Yes	Yes
Patenting status, up to 1895	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.034	0.028	0.029	0.039	0.038	0.036
Fill value c	0.179	0.179	0.179	0.416	0.416	0.416

Notes: Columns (1) and (4) restrict the sample to firms used in the lower-bound estimation; column (2) and (5) include all firms from the Budapest exhibition; column (3) and (6) restrict the sample to firms used in the upper-bound estimation. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Taken together, the evidence points one way. Innovation rose, but only among firms that began exporting, and it did not move firms toward the specific technologies displayed at the Paris trade fair. Innovation of Paris exhibitors was a consequence of market access rather than an independent channel of technology transfer and learning between exhibitors.

Two caveats bound the claim. First, patents capture codified, legally protected innovation, so process improvements, design changes, and organizational learning that firms cannot patent would not appear in any of our three tests: We can rule out technology absorption that reached the patent office, not all learning. Second, the tests separate learning at the trade fair from innovation through the export margin, but they cannot separate the two channels that both operate through the export margin: a firm that innovates because a larger market pays for it and a firm that innovates because demanding foreign buyers require it look identical here.

8 Consequences

Ultimately, for policymakers, exporting is a means to an end rather than an objective in itself. Governments use export promotion instruments because they want firms to survive

and grow. In this section, we ask whether participation in the Paris trade fair delivered on these goals and how persistent its effects were. We first study whether participation caused firms to expand, measured by employment growth, and whether it raised the probability of survival. We then extend the time horizon and study whether the export effect persisted until 1909, nine years after the exhibition. Some of the firm names in our sample are still well known in Hungary. This motivated us to ask in the final section whether any trace of the fair is more systematically visible today, more than a century and several political and economic disruptions later.

8.1 Employment growth

In the standard heterogeneous-firm framework (Melitz, 2003), firms that gain access to export markets expand production to serve the additional demand, and this scale effect increases employment. If firms face frictions to growth, such as investment or financing constraints, they may instead meet foreign demand by reallocating existing capacity away from domestic customers, in which case employment would be unchanged. The employment response therefore tells us which adjustment margin firms used. Employment is also a policy objective in its own right, which makes the magnitude of interest independent of what it reveals about the mechanism.

We observe employment in both 1896 and 1904 for a subset of firms.[28] For this subset, we find that participation at the Paris trade fair raised employment by between 17.7 and 27.1 percent, with a within estimate of 25.9 percent.[29] Over the eight years between the two measurements, this corresponds to annual additional growth of between 2.1 and 3.0 percent,[30] The lower bound remains statistically significant despite the smaller sample. The estimates follow the ordering stated in Lemma 4, although the within estimate lies very close to the upper bound.[31]

These effects speak against a pure reallocation of existing capacity: Paris participants did not simply redirect output from domestic to foreign customers, but expanded. This contrasts with Atkin et al. (2017), where Egyptian rug producers given access to foreign buyers increased output prices by 43 percent and improved quality, but produced less and hired no additional workers, expanding labor input only through a small increase in hours. Large trade fairs therefore appear to differ from targeted buyer-matching interventions in

[28] Out of the 3,614 firms, 644 (18%) reported employment information both in 1896 and 1904. The availability of information also raises with Budapest exhibition award classes.

[29] Percentage changes are computed as $\exp(\hat{\beta}) - 1$.

[30] Annualized as $\exp(\hat{\beta}/8) - 1$.

[31] Appendix Table D.9 replicates Table 10 on the common sample.

the kind of adjustment they induce.

Table 10: The effect of the Paris exhibition on employment growth

	ln(workers), 1904		
	(1)	(2)	(3)
Paris exhibitor	0.163**	0.230***	0.240***
	(0.078)	(0.064)	(0.062)
Pre-treatment outcome	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper
Sample	Lower	Within	Upper
Observations	538	644	601
R-squared	0.175	0.188	0.196
Difference from within estimate	0.067		0.011
	(0.056)		(0.045)

Notes: Each column estimates equation (1) with the change in log employment between 1896 and 1904 as the outcome, on the subset of firms for which employment is observed in both years. Column (1) restricts the sample to firms used in the lower-bound estimation; column (2) includes all firms from the Budapest exhibition; column (3) restricts the sample to firms used in the upper-bound estimation. The pre-treatment outcome is ln(workers) in 1896. The regressions calculating the difference from the *within* estimate are described in Appendix D.3. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

8.2 Firm survival

To examine how participation affected firm survival, we use two alternative measures. The first is based on firm exits between 1896 and 1904, recorded in the exit reports of the *Central Gazette of Hungarian Firms*. Since no complete firm registry exists for this period, however, we cannot be certain that these records are comprehensive. We therefore complement them with a second, independent source: the 1906–1907 *Magyar Gyárak és Iparvállalatok Címtára* (Directory of Hungarian Factories and Industrial Enterprises), which lists a subset of firms, likely the larger and more mechanized establishments. Appearance in the directory requires survival at least through 1906–1907, but it also requires reaching a size threshold, so it is best read as a joint measure of survival and size rather than of survival alone.

Columns (1) to (3) of Table 11 use the first measure as outcome, an indicator for the 122 firms (3.3 percent of the sample) that appear in the exit reports. Paris participation reduced the probability of exit, with estimates ranging from 1.9 to 3.7 percentage points and all three statistically significant. For an outcome where a beneficial effect is negative, the conservative

estimate is the one closest to zero, here the 1.9 percentage points in column (3). Against a baseline exit rate of 3.3 percent, even this conservative estimate implies that participation cut the probability of exit by more than half. Columns (4) to (6) use the second measure, appearance in the directory. The findings are consistent: Paris participation raised the probability of being listed among the larger establishments, with estimates ranging from 3.9 to 26.8 percentage points, although the lower bound is not statistically distinguishable from zero.

Overall, the finding that Paris participation lowered exit and raised the likelihood of surviving as a larger establishment is consistent with the well-documented positive association between exporting and survival (Bernard and Jensen, 1999), for instance because access to export markets lifts firm revenue above the exit threshold (Melitz, 2003).

Table 11: The effect of the Paris exhibition on firm exit and directory appearance

	Firm exit, 1896–1904			Directory appearance, 1906		
	(1)	(2)	(3)	(4)	(5)	(6)
	Lower bound	Within	Upper bound	Lower bound	Within	Upper bound
Paris exhibitor	−0.037***	−0.025***	−0.019***	0.039	0.138***	0.268***
	(0.012)	(0.008)	(0.007)	(0.037)	(0.030)	(0.033)
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,742	3,736	3,615	2,650	3,614	3,498
R-squared	0.037	0.029	0.028	0.217	0.226	0.222

Notes: Each column estimates equation (1). Columns (1) to (3) add the 122 firms that exited between 1896 and 1904 to the sample and use an indicator for exit as the outcome. Columns (4) to (6) use appearance in the 1906–1907 Directory of Hungarian Factories and Industrial Enterprises as the outcome. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

8.3 Longer run: extra-regional trade by 1909

A further issue of the Hungarian National Museum of Commerce register, for 1909, allows us to re-estimate the treatment effects on extra-regional trade for the longer run, nine years after the Paris trade fair.

The treatment effect is still positive and bounded between 5.0 and 17.9 percentage points, with a within estimate of 11.3 (Table 12), compared to 7.6 and 20.6 percentage points for the same outcome measured in 1904 (Table 4). Both causal bounds are thus smaller after nine years than after four, and the lower bound is now significant only at the 10 percent level. Nonetheless, the causal bounds remain entirely positive. Relative to the 1898 baseline extra-regional trade share of 25 percent, the effects fall from between 30 and 82 percent in

1904 to between 20 and 72 percent by 1909.[32]

The effect of the exhibition, therefore, attenuates but does not disappear within a decade. This is what one would expect if the fair worked through matching: a match formed in 1900 generates trade for as long as the relationship lasts, and relationships end. Nine years on, appoximately two thirds of the initial effect remains for the lower bound example, suggesting that the relationships formed at the fair were relatively durable, or that firms that lost a buyer were often able to find another.

Table 12: The effect of the Paris exhibition on extra-regional trade in 1909

	Extra-regional trade, 1909		
	(1)	(2)	(3)
	Lower bound	Within	Upper bound
Paris exhibitor	0.050*	0.113***	0.179***
	(0.029)	(0.023)	(0.028)
Pre-treatment outcome	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper
Sample	Lower	Within	Upper
Observations	2,650	3,614	3,498
R-squared	0.469	0.476	0.481

Notes: Each column estimates equation (1). The pre-treatment outcome is extra-regional trade in 1898. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

8.4 Very long run: survival into the present

We now extend the time horizon to the present. This evidence should be read as suggestive rather than conclusive: the outcome is rare, the coding is necessarily judgmental, and the Hungarian economy has gone through upheavals over the past century, including the First World War, the Treaty of Trianon, the Second World War, the nationalizations of 1948, and the 1990 transition. These caveats notwithstanding, we report the outcomes because the raw pattern is striking.

Strict legal continuity is close to impossible in our setting. Hungary lost about two thirds of its pre-war territory in 1920, so a non-trivial share of our 1896 firms ended up in successor states, and the nationalizations of 1948-1949 eliminated private firm ownership altogether. A firm that is recognizable today is therefore never the same legal entity as in 1896: it has been re-founded, re-privatized, or is a different firm using the historical brand. For this

[32]Appendix Table D.10 replicates the estimates on the common sample.

reason, we use two alternative definitions of firm continuity that differ on stringency. *Firm survival* is our strictest measure and identifies firms that operate today in the same line of business and under a recognizably continuous name, even where continuity was interrupted by nationalization. These cases are individually well documented. *Name survival* is broader measure and also counts identifiable successor companies and cases in which only the brand survives.

Table 13 reports the raw data. Under the stricter definition we document eight surviving firms out of the 3,614 in our sample.[33] Six of them exhibited in Paris: the ceramics manufacturers *Zsolnay* and *Herend*, the Budapest distiller of Unicum bitter, *Zwack*, the winter salami maker *Pick* of Szeged, the Budapest confectioner *Gerbeaud*, and the furniture maker *Thonet*. Two did not: the printing house *Kner Izidor* and the confectioner *Auguszt*.

In the raw data, survival is more than ten times (1.1 versus 0.1 percent) more likely among the 533 Paris exhibitors than among the 3,081 firms that stayed at home. If we allow for name survival, the gap is smaller but still large: the names of 5.3 percent of Paris exhibitors survive, compared to 0.8 percent of non-Paris exhibitors. Both differences are statistically significant according to Fisher exact tests.

Table 13: Long-run survival, by Paris exhibition status

	Paris exhibitors		Non-exhibitors		
	Count	Share	Count	Share	p-value
Firm survival	6	0.011	2	0.001	0.000
Name survival	28	0.053	25	0.008	0.000
Number of firms	533		3,081		

Notes: Sample of 3,614 firms from the 1896 Budapest exhibition. The last column reports the two-sided p-value of a Fisher exact test of independence between exhibition status and survival.

Table 14 uses the two long-run indicators as outcomes in our research design. With only eight events under the stricter definition, these estimates are inevitably imprecise, and we see them as descriptive rather than as conclusive evidence. The causal bounds range from 0.8 to 1.3 percentage points for firm survival and from 0.2 to 4.7 percentage points for name survival. Both lower bounds are statistically insignificant, but the magnitudes are large relative to the corresponding non-exhibitor baselines of 0.06 and 0.8 percent.[34]

[33] See Appendix G.12 for details on these firms.

[34] Note that the survival sample conditions on firms that had not exited by 1904. This is conditioning on a post-treatment outcome, but its direction works against us, as they measure the effect on survival to the

Table 14: Treatment effects on long-run survival

	Firm survival			Name survival		
	(1)	(2)	(3)	(4)	(5)	(6)
	Lower	Within	Upper	Lower	Within	Upper
Paris exhibitor	0.008	0.008**	0.013**	0.002	0.019***	0.047***
	(0.005)	(0.004)	(0.006)	(0.008)	(0.007)	(0.011)
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.035	0.058	0.048	0.080	0.121	0.112
Mean, non-exhibitors		0.001			0.008	

Notes: Outcomes are defined in Section 8.4. Columns (1) and (4) restrict the sample to firms used in the lower-bound estimation; columns (2) and (5) include all firms from the Budapest exhibition; columns (3) and (6) restrict the sample to firms used in the upper-bound estimation. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

9 Conclusion

This paper provides causal evidence that a trade fair can promote exports by reducing search and matching frictions. Using the 1900 Paris Exhibition as a natural experiment, we show that participation raised Hungarian firms' probability of extra-AHM trade by 4.2 to 18.3 percentage points and of extra-regional trade by 7.6 to 20.6 points, against baselines of 15 and 25 percent. Participants also grew faster, by 2.1 to 3 percent of annual employment growth, and were 4 to 9 percentage points more likely to patent over the following five years. The export effect was still visible in 1909, nine years after the event. The commercial relationships formed in Paris seem to have had lasting value.

A key contribution is to show which firms benefit the most from a trade fair. The gains mainly accrued to firms that were already exporting: the fair helped them sustain and expand foreign trade, and did not turn non-exporters into exporters. Returns were highest where the buyer side was thick and domestic rivals were few, so the composition of the exhibitor pool mattered as much as attendance itself. Awards worked in the same direction. A Paris medal was worth the most to firms facing many potential buyers, so certification seems to have operated through matching and not through brand value alone. Innovation fits this pattern as well. Patenting rose, but it rose together with exporting rather than beside it, and

present *in addition to* the survival effect already visible by 1904 in Table 11.

firms did not shift the content of their patents toward the technology on display. At Paris, innovation was a consequence of market access rather than a separate channel of technology transfer.

The paper also makes a methodological contribution. We use predetermined export potential and the constraints on the selection committee to bound the treatment effect from two sides, comparing treated firms with controls that are close from above and close from below. The design needs no assumption about how firms were chosen within an award class, where selection may be positive, negative or absent, since nothing the committee did moved a firm from one class to another. Compared with standard partial identification bounds, ours are substantially tighter and exclude zero, and hence they can inform a policy decision. Matching on observables does not solve the problem either: it reproduces the within-class comparison rather than correcting it, and our balancing tests indicate that this comparison is upward biased. When randomization is infeasible, institutional detail of this kind can substitute for it, provided two things are present: a coarse ranking of the outcome-relevant characteristic fixed before treatment, and a selection rule imperfect enough to leave treated and untreated firms inside the same bin.

We should emphasize what we (and several similar exercises) have and have not measured. Our estimates are the *private* return to selected participation, the gain to a firm sent to Paris while other similar firms stayed home. This is not the *aggregate* return to expanding participation because the fair is a congestible platform. Sending many more similar firms would change the very congestion conditions that our heterogeneity estimates show drive the effect.

Despite resting on a historical episode, our findings carry implications for contemporary export promotion policy, where public spending on trade fair subsidies is substantial and causal evidence on their effectiveness remains thin. Simply maximizing participation may be counterproductive, since sending too many similar firms dilutes the benefit to each. Export promotion agencies might limit the number of supported firms per product category and prioritize sectors where a large international buyer base can be reached. More broadly, our findings favor well-organized large events that concentrate many potential buyers with relatively few supported firms per niche. The lesson from our episode is to send firms that already have a track record abroad and a chance to stand out.

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A Proofs

Proof of Lemma 1.

$$\begin{aligned}\beta^a_{lower} &= E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a-1] \\ &= \underbrace{E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=1, A=a]}_{=\tau^a} \\ &\quad + \underbrace{E[Y(0) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a-1]}_{\leq 0},\end{aligned}$$

$$\begin{aligned}\beta^a_{upper} &= E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a+1] \\ &= \underbrace{E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=1, A=a]}_{=\tau^a} \\ &\quad + \underbrace{E[Y(0) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a+1]}_{\geq 0}.\end{aligned}$$

Both signs follow from Assumption 1. □

Proof of Lemma 2.

$$\begin{aligned}\beta^a_{within} &= E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a] \\ &= \underbrace{E[Y(1) \mid D=1, A=a] - E[Y(0) \mid D=1, A=a]}_{=\tau^a} \\ &\quad + \underbrace{E[Y(0) \mid D=1, A=a] - E[Y(0) \mid D=0, A=a]}_{=0},\end{aligned}$$

where the second term is zero by Assumption 2. □

Proof of Lemma 3. Follows directly from Assumption 1 with $d = d'$. □

Proof of Lemma 4. The three comparisons share the term $E[Y(1) \mid D=1, A=a]$ and differ only in the award class supplying the control firms, so

$$\begin{aligned}\beta^a_{within} - \beta^a_{lower} &= E[Y(0) \mid D=0, A=a-1] - E[Y(0) \mid D=0, A=a] \;\geq\; 0, \\ \beta^a_{upper} - \beta^a_{within} &= E[Y(0) \mid D=0, A=a] - E[Y(0) \mid D=0, A=a+1] \;\geq\; 0,\end{aligned}$$

where both inequalities follow from Assumption 1 with $d = d' = 0$. □

B Manski bounds

This appendix reviews Manski's partial identification approach and shows how our award-based bounds tighten the Manski bounds.

B.1 Definition of Manski bounds

Our object of interest is the average treatment effect on the treated (ATT),

$$\text{ATT} = \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1], \tag{B.1}$$

where we observe only

$$Y = DY(1) + (1 - D)Y(0), \quad D \in \{0, 1\},$$

and we do not impose random assignment, selection-on-observables, instrumental variables, or other identifying assumptions. As a result, the counterfactual mean outcome for treated units,

$$\mathbb{E}[Y(0) \mid D = 1],$$

is not identified from the data. In our setting, Y is a dummy outcome (e.g., extra-AHM trade), so $Y \in [0, 1]$.

The key insight of Manski (2003) is that even without additional identifying assumptions, one can bound treatment effects using only the fact that outcomes are bounded.[35] For a binary outcome, the natural bounds are 0 and 1.

Assumption B.1 (Bounded outcomes)**.** *The potential outcomes are bounded in* $[0, 1]$*:*

$$0 \leq Y_i(0) \leq 1 \quad \textit{and} \quad 0 \leq Y_i(1) \leq 1 \qquad \textit{for all } i.$$

The Manski bounds for the ATT follow immediately by combining assumption B.1 with equation (B.1). The lower bound is obtained by setting the counterfactual to its largest feasible value (1), while the upper bound sets it to its smallest feasible value (0).

Lemma B.1 (Manski bounds)**.** *Define the upper and lower bounds on the ATT as*

$$\overline{ATT}_{Manski} = \mathbb{E}[Y(1) \mid D = 1] - 0,$$
$$\underline{ATT}_{Manski} = \mathbb{E}[Y(1) \mid D = 1] - 1.$$

Under assumption B.1,

$$\underline{ATT}_{Manski} \leq ATT \leq \overline{ATT}_{Manski}.$$

Proof.

$$\mathbb{E}[Y(1) \mid D = 1] - 1 \leq \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1] \leq \mathbb{E}[Y(1) \mid D = 1] - 0.$$

□

[35] Manski (2003) typically presents the argument for the ATE. Since our focus is the ATT, we state the bounds for $\mathbb{E}[Y(0) \mid D = 1]$ and hence for the ATT.

That is, any value of the ATT outside $[\underline{\text{ATT}}_{\text{Manski}}, \overline{\text{ATT}}_{\text{Manski}}]$ is incompatible with the data. These bounds are sharp. Because these assumptions are extremely weak, the resulting bounds can be wide. With a binary outcome, the width equals 1, so the ATT bounds necessarily include zero. This is often too coarse for inference, for instance when the goal is to rule out null effects.

B.2 Tightening the Manski bounds: monotone treatment response (MTR) and monotone treatment selection (MTS)

Two common additional restrictions are the following. First, *monotone treatment selection (MTS)* assumes that treated units have weakly higher counterfactual outcomes than untreated units.

Assumption B.2 (Monotone Treatment Selection (MTS))**.**

$$\mathbb{E}[Y(0) \mid D = 0] \leq \mathbb{E}[Y(0) \mid D = 1].$$

Second, *monotone treatment response (MTR)* assumes that treatment weakly improves outcomes for each treated unit.

Assumption B.3 (Monotone Treatment Response (MTR))**.**

$$\mathbb{E}[Y(0) \mid D = 1] \leq \mathbb{E}[Y(1) \mid D = 1].$$

Together, these restrictions tighten the identified set for the ATT:

Lemma B.2 (Manski bounds with MTS and MTR)**.** *Define the upper bound using MTS and the lower bound using MTR for the counterfactuals:*

$$\overline{ATT}_{MTS} = \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0],$$
$$\underline{ATT}_{MTR} = \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(1) \mid D = 1] = 0.$$

Under assumptions B.2 and B.3,

$$\underline{ATT}_{MTR} \leq ATT \leq \overline{ATT}_{MTS}.$$

The upper bound under MTS, $\overline{\text{ATT}}_{\text{MTS}}$, is often referred to as the *naive* comparison, since it effectively compares treated outcomes to untreated outcomes without accounting for selection.

B.3 Comparison to our causal bounds

Our approach extends the Manski approach by invoking Assumption 1, which restricts two subgroup comparisons across adjacent award classes:

$$\mathbb{E}[Y(0) \mid D = 0, A = a - 1] \geq \mathbb{E}[Y(0) \mid D = 1, A = a]$$

for the lower bound, and

$$\mathbb{E}[Y(0) \mid D = 1, A = a] \geq \mathbb{E}[Y(0) \mid D = 0, A = a + 1]$$

for the upper bound. To see how these deliver bounds on the ATT, rewrite it by conditioning on award class A:

$$\begin{aligned} \text{ATT} &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1] \\ &= \sum_{a \in \mathcal{A}} \Big(\mathbb{E}[Y(1) \mid D = 1, A = a] - \mathbb{E}[Y(0) \mid D = 1, A = a] \Big) \Pr(A = a \mid D = 1), \quad \text{(B.2)} \end{aligned}$$

where $\Pr(A = a \mid D = 1)$ is the share of treated units in award class a.

Lemma B.3 (Our causal bounds)**.** *Define the lower bound as*

$$\underline{ATT} = \sum_{a \in \mathcal{A} \setminus \{1\}} \Big(\mathbb{E}[Y(1) \mid D = 1, A = a] - \mathbb{E}[Y(0) \mid D = 0, A = a - 1] \Big) \widetilde{\Pr}(A = a \mid D = 1),$$

and the upper bound as

$$\overline{ATT} = \sum_{a \in \mathcal{A} \setminus \{5\}} \Big(\mathbb{E}[Y(1) \mid D = 1, A = a] - \mathbb{E}[Y(0) \mid D = 0, A = a + 1] \Big) \widehat{\Pr}(A = a \mid D = 1),$$

where

$$\widetilde{\Pr}(A = a \mid D = 1) \equiv \Pr(A = a \mid D = 1, A \neq 1) = \frac{\Pr(A = a \mid D = 1)}{\Pr(A \neq 1 \mid D = 1)}$$

and

$$\widehat{\Pr}(A = a \mid D = 1) \equiv \Pr(A = a \mid D = 1, A \neq 5) = \frac{\Pr(A = a \mid D = 1)}{\Pr(A \neq 5 \mid D = 1)}$$

rescale the weights to sum to one over the award classes for which the respective comparison is defined. Under Assumption 1, and assuming that the treatment effect for the top award class is at least as large as $\underline{ATT}$ and the treatment effect for the class without an award is no larger than $\overline{ATT}$,

$$\underline{ATT} \leq ATT \leq \overline{ATT}.$$

Both bounds differ from Manski, MTR, and MTS in that they use the counterfactual of a specific group of untreated firms rather than the extreme feasible value or the full untreated population. Because each construction relies on the adjacent award class, the lower bound excludes the top award class and the upper bound excludes the class without an award. The two bounds are therefore constructed on different sets of treated firms, and the extrapolation conditions in the lemma are what allow us to use them as bounds on the ATT for all treated firms. In the main text we also report all three comparisons on the common sample of award classes 2 to 4, for which no extrapolation is needed.

Assumption 2 imposes the MTS restriction in Assumption B.2 as an equality and only within award classes, which is why β_{within} point identifies the treatment effect under it rather than bounding it.

B.4 Manski MIV bounds

Another refinement in the literature is to use an additional covariate that is believed to be related to potential outcomes as a Manski *Monotone Instrumental Variable (MIV)* (Manski and Pepper, 2000). A natural application in our setting is to use award categories A_i as the Manski MIV. Recall that $A_i \in \mathcal{A} = \{1, 2, 3, 4, 5\}$, where $A_i = 1$ indicates the highest export potential.[36] It is natural to assume that firms in better award categories have (weakly) higher export potential:

Assumption B.4 (Manski MIV)**.** *If $a_1 \leq a_2$, then*

$$\mathbb{E}[Y(0) \mid A = a_1] \geq \mathbb{E}[Y(0) \mid A = a_2].$$

The following lemma illustrates how this assumption can be used to construct bounds for the counterfactual outcomes, which can then be used to construct bounds for the treatment effect on the treated ATT:[37]

Lemma B.4 (Manski MIV bounds)**.** *Assume that outcomes are bounded as in assumption B.1, and assume $\Pr(D = 1 \mid A = a) > 0$ for all $a \in \mathcal{A}$ with $\Pr(A = a \mid D = 1) > 0$. Define the award-specific bounds on the counterfactual outcome for the treated as*

$$\overline{\mathbb{E}}[Y(0) \mid D = 1, A = a] \equiv \min \Bigg\{ 1,\ \frac{1}{\Pr(D = 1 \mid A = a)} \cdot \Bigg[\inf_{\tilde{a} \leq a} \Big(\Pr(D = 1 \mid A = \tilde{a}) + \mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) - \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a) \Big) \Bigg] \Bigg\},$$

$$\underline{\mathbb{E}}[Y(0) \mid D = 1, A = a] \equiv \max \Bigg\{ 0,\ \frac{1}{\Pr(D = 1 \mid A = a)} \cdot \Bigg[\sup_{\tilde{a} \geq a} \Big(\mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) - \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a) \Big) \Bigg] \Bigg\}.$$

Define

$$\overline{ATT}_{IV} \equiv \sum_{a \in \mathcal{A}} \Big(\mathbb{E}[Y(1) \mid D = 1, A = a] - \underline{\mathbb{E}}[Y(0) \mid D = 1, A = a] \Big) \Pr(A = a \mid D = 1),$$

$$\underline{ATT}_{IV} \equiv \sum_{a \in \mathcal{A}} \Big(\mathbb{E}[Y(1) \mid D = 1, A = a] - \overline{\mathbb{E}}[Y(0) \mid D = 1, A = a] \Big) \Pr(A = a \mid D = 1).$$

[36] Notice that the ordering of covariates differs from the classical approach in Manski and Pepper (2000): in our setting, a smaller a corresponds to higher potential outcomes, for example, Gold has a smaller a than Silver.

[37] It is natural to assume that firms in better award classes have weakly higher export potential. Manski and Pepper (2000) state the restriction for both potential outcomes; since we bound the ATT, only the restriction on $Y(0)$ is used, so we state it for $Y(0)$ alone.

Under assumption B.4,

$$\underline{ATT}_{IV} \leq ATT \leq \overline{ATT}_{IV}.$$

Proof. By the law of total expectation,

$$\mathbb{E}[Y(0) \mid A = a] = \mathbb{E}[Y(0) \mid D = 1, A = a] \Pr(D = 1 \mid A = a) + \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a). \tag{B.3}$$

Upper bound on the counterfactual. In this expression, only $\mathbb{E}[Y(0) \mid D = 1, A = a]$ is unobserved, but since $Y(0) \leq 1$, we have $\mathbb{E}[Y(0) \mid D = 1, A = a] \leq 1$. Therefore:

$$\mathbb{E}[Y(0) \mid A = a] \leq \Pr(D = 1 \mid A = a) + \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a). \tag{B.4}$$

Assumption B.4 implies that for all $\tilde{a} \leq a$,

$$\mathbb{E}[Y(0) \mid A = a] \leq \mathbb{E}[Y(0) \mid A = \tilde{a}].$$

Therefore,

$$\begin{aligned} \mathbb{E}[Y(0) \mid A = a] &\leq \inf_{\tilde{a} \leq a} \mathbb{E}[Y(0) \mid A = \tilde{a}] \\ &\leq \inf_{\tilde{a} \leq a} \Big(\Pr(D = 1 \mid A = \tilde{a}) + \mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) \Big), \end{aligned}$$

where the last inequality applies equation (B.4) to all award classes $\tilde{a} \leq a$.

Substituting expression (B.3) into the left-hand side of this inequality and rearranging for $\mathbb{E}[Y(0) \mid D = 1, A = a]$ gives

$$\begin{aligned} \mathbb{E}[Y(0) \mid D = 1, A = a] \leq \frac{1}{\Pr(D = 1 \mid A = a)} \cdot & \\ \Bigg[\inf_{\tilde{a} \leq a} \Big(\Pr(D = 1 \mid A = \tilde{a}) + \mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) \Big) & \\ - \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a) \Bigg]. & \end{aligned}$$

Since $Y(0) \leq 1$, also $\mathbb{E}[Y(0) \mid D = 1, A = a] \leq 1$, so taking the minimum with 1 further tightens the bound.

Lower bound on the counterfactual. Applying the lower bound on the outcome $Y(0) \geq 0$ on equation (B.3), we get $\mathbb{E}[Y(0) \mid D = 1, A = a] \geq 0$. Therefore:

$$\mathbb{E}[Y(0) \mid A = a] \geq \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a). \tag{B.5}$$

Assumption B.4 implies that for all $\tilde{a} \geq a$,

$$\mathbb{E}[Y(0) \mid A = a] \geq \mathbb{E}[Y(0) \mid A = \tilde{a}].$$

Therefore,

$$\begin{aligned}\mathbb{E}[Y(0) \mid A = a] &\geq \sup_{\tilde{a} \geq a} \mathbb{E}[Y(0) \mid A = \tilde{a}] \\ &\geq \sup_{\tilde{a} \geq a} \Big(\mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) \Big),\end{aligned}$$

where the last inequality applies equation (B.5) to all award classes $\tilde{a} \geq a$.

Substituting expression (B.3) into the left-hand side of this inequality and rearranging for $\mathbb{E}[Y(0) \mid D = 1, A = a]$ gives

$$\begin{aligned}\mathbb{E}[Y(0) \mid D = 1, A = a] \geq \frac{1}{\Pr(D = 1 \mid A = a)} \cdot & \\ \Bigg[\sup_{\tilde{a} \geq a} \Big(\mathbb{E}[Y(0) \mid D = 0, A = \tilde{a}] \Pr(D = 0 \mid A = \tilde{a}) \Big) & \\ - \mathbb{E}[Y(0) \mid D = 0, A = a] \Pr(D = 0 \mid A = a) \Bigg]. &\end{aligned}$$

Since $Y(0) \geq 0$, also $\mathbb{E}[Y(0) \mid D = 1, A = a] \geq 0$, so taking the maximum with 0 further tightens the bound.

Combining both inequalities yields, for each a,

$$\underline{\mathbb{E}}[Y(0) \mid D = 1, A = a] \leq \mathbb{E}[Y(0) \mid D = 1, A = a] \leq \overline{\mathbb{E}}[Y(0) \mid D = 1, A = a].$$

Subtracting these inequalities from $\mathbb{E}[Y(1) \mid D = 1, A = a]$, multiplying by $\Pr(A = a \mid D = 1)$, and summing over $a \in \mathcal{A}$ (using again (B.2)) gives $\underline{\text{ATT}}_{\text{IV}} \leq \text{ATT} \leq \overline{\text{ATT}}_{\text{IV}}$. □

Relative to the unconditional Manski bounds, the MIV restriction uses the award ordering to tighten the identified set by exploiting information across award classes. The upper bound on $\mathbb{E}[Y(0) \mid D = 1, A = a]$ compares award class a to all higher-potential classes $\tilde{a} \leq a$, and the lower bound to all lower-potential classes $\tilde{a} \geq a$.

Assumption 1 exploits the same award ordering but restricts different objects. Assumption B.4 orders the mixture of treated and untreated firms across award classes, that is, $\mathbb{E}[Y(0) \mid A = a]$. Assumption 1 instead orders the subgroups that the bounds actually compare: untreated firms in class $a - 1$ against treated firms in class a for the lower bound, and treated firms in class a against untreated firms in class $a + 1$ for the upper bound. Neither assumption implies the other. The MIV restriction is weaker in that it says nothing about how firms are selected within a class, but it is also less informative, because the mixture it orders contains the unobserved counterfactual outcomes of the treated firms, which is precisely the object to be bounded. Our assumption orders only observed subgroup means against the counterfactual mean of the treated in an adjacent class, which is why it delivers tighter bounds in our setting.

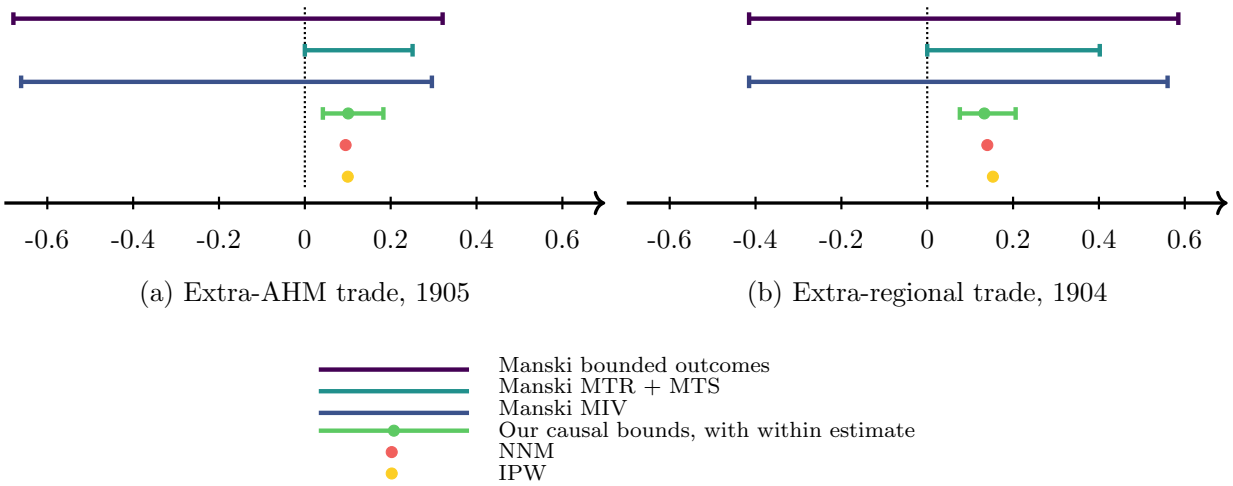
B.5 Application of Manski bounds and matching estimators to our setting

In this section we compare the bounds generated by each approach in our setting, illustrated in Figure B.1.

We begin with the weakest assumption: that outcomes are bounded. For a binary outcome such as extra-AHM trade, $Y \in [0, 1]$. The Manski bounds set the unobserved counterfactual outcomes to their extreme feasible values: the lower bound assumes all treated units would have exported without treatment, while the upper bound assumes none would have. The first line in Figure B.1 reports the resulting bounds: $[-0.435, 0.565]$ for extra-regional trade and $[-0.688, 0.311]$ for extra-AHM trade. With binary outcomes, these bounds always have width 1 and necessarily include zero, making them too wide for inference.

Researchers often impose two additional restrictions to tighten these bounds. *Monotone treatment selection (MTS)* assumes that treated units have weakly higher counterfactual outcomes than untreated units, while *monotone treatment response (MTR)* assumes that treatment weakly improves outcomes for each unit. The second line shows how MTR and MTS narrow the bounds: the lower bound becomes 0 by construction, and the upper bound reduces to the naïve comparison $\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$, which ignores selection. The resulting intervals are $[0, 0.390]$ for extra-regional trade and $[0, 0.242]$ for extra-AHM trade. While substantially tighter, these bounds still include zero by construction and therefore cannot rule out null effects.

The third line applies an alternative refinement: using the award categories as a *Monotone Instrumental Variable* (MIV) in the sense of Manski and Pepper (2000). Despite exploiting the same covariate that our approach uses, the MIV bounds are wide, tightening the bounded-outcomes intervals only marginally: $[-0.435, 0.543]$ for extra-regional trade and $[-0.667, 0.291]$ for extra-AHM trade.



(a) Extra-AHM trade, 1905

(b) Extra-regional trade, 1904

Figure B.1: Comparison to Manski bounds and matching estimators

The fourth line applies Assumption 1 within the same partial-identification logic. Relative to MTS, which orders treated and untreated firms globally, our assumption orders subgroups across adjacent award classes. Relative to MIV, which orders the mixture of

treated and untreated firms in an award class against all higher or lower classes, both of our bounds restrict the comparison to untreated firms in the adjacent class alone. Because our assumption neither nests within nor is nested by MTR, MTS, or MIV, the resulting identified set need not be contained in these alternative bounds. Empirically, however, our causal bounds are substantially tighter, excluding zero in both outcomes. The dot marks β_{within}, which lies inside the bounds as Lemma 4 requires. Our causal bounds as well as the within-estimate come from the specification in Section 4.3, which conditions on the baseline outcome and product group, while the Manski, MTS, and MIV bounds are unconditional, so part of the tightening comes from the conditioning.

The last two lines report point estimates from two conventional matching approaches that researchers commonly use to control for selection on observables. We treat the awards as unobserved and instead predict Paris participation using baseline firm characteristics alone (see Appendix Table B.1). Past extra-AHM trade status, extra-regional trade status, reporting horsepower information, and being a joint-stock company are significant predictors of participation. We then implement nearest-neighbor matching (NNM, with 5 neighbors) and inverse probability weighting (IPW) based on these predicted propensities. Both deliver estimates at or above β_{within}, but the balancing checks in Section 5.2 suggested that the latter is upward biased. Matching on observables therefore reproduces the within-class comparison rather than correcting it, because the characteristics we observe do not capture what the Budapest award jury saw. This underscores the value of exploiting the award classes as a predetermined proxy for unobserved export potential.

Table B.1: First stage: nearest neighbor matching and propensity score estimation

	(1) Paris exhibitor
Extra-AHM trade (1896)	0.319**
	(0.151)
Extra-regional trade (1898)	1.108***
	(0.169)
Horsepower info available	0.619***
	(0.167)
Worker information available	0.161
	(0.178)
Patent indicator	-0.271
	(0.169)
Joint-stock company	0.920***
	(0.214)
Output information available	−0.236
	(0.159)
Multi-location firm	0.227
	(0.186)
Observations	3,565
Product FE	Yes

Notes: Logit estimation. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

C Appendix figures

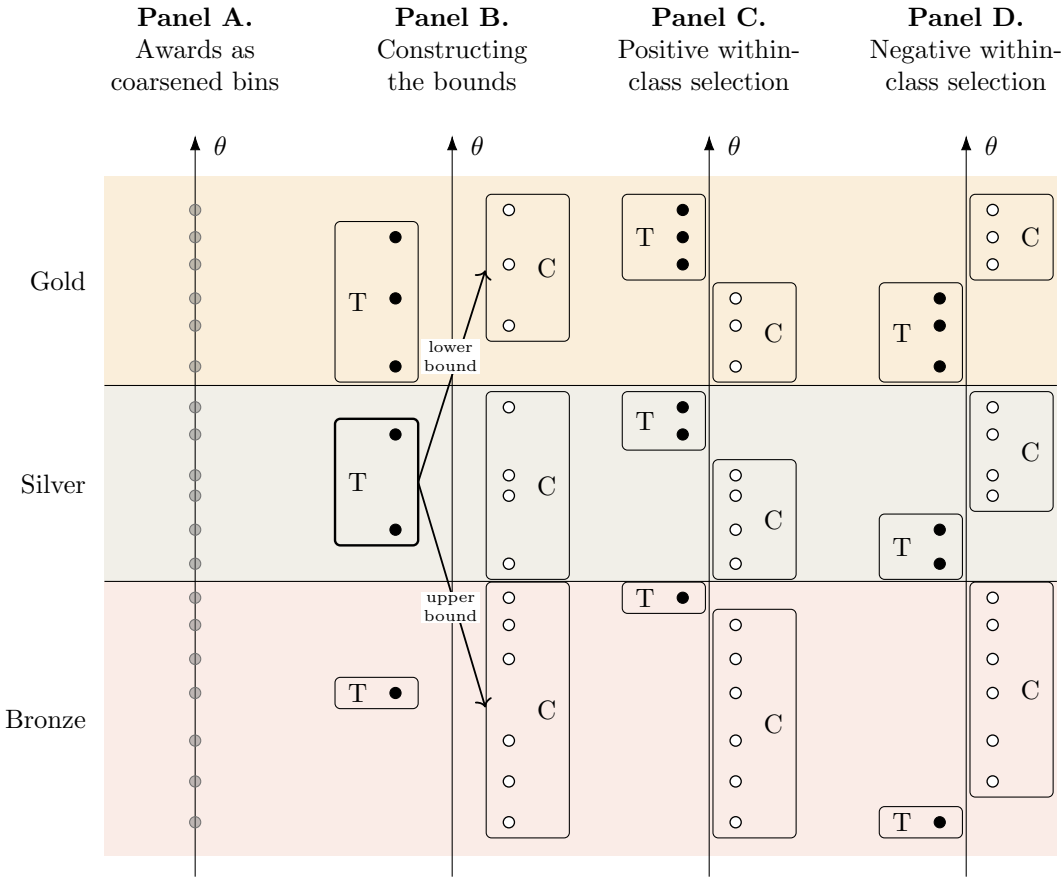


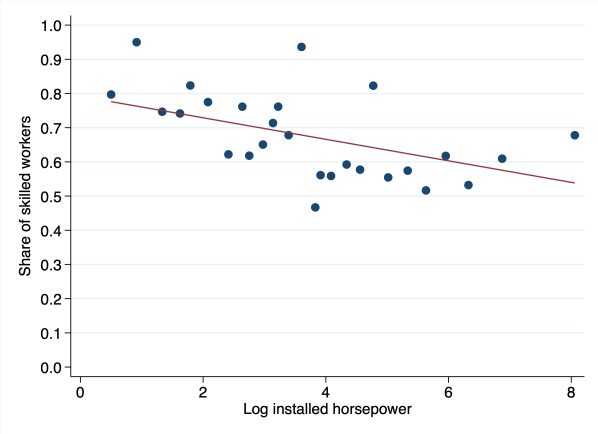
Figure C.1: Using award classes to construct causal bounds: example

Notes: The vertical axis shows unobserved export potential θ; shading indicates award classes. Circles denote firms, grouped into Paris participants (T) and non-participants (C) within each award class.

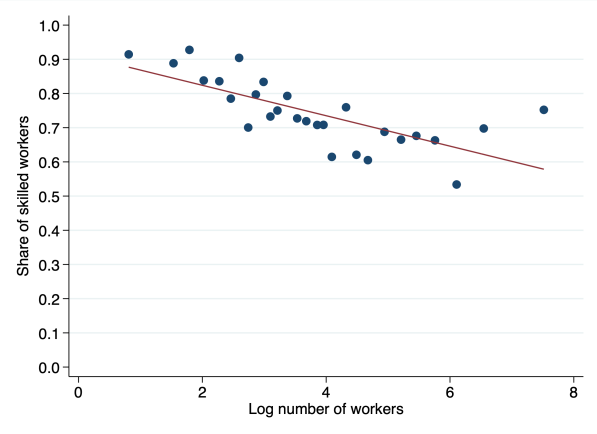


Figure C.2: Balancing tests on pre-treatment continuous variables lower bound, within-estimate, and upper bound

Notes: The figure plots the coefficient from estimating equation (1) with pre-treatment continuous variables as dependent variables, using the respective comparison groups.



(a) Skill share and mechanization



(b) Skill share and employment

Figure C.3: Relationship between skill share and mechanization or employment

Notes: Each dot represents the mean share of skilled workers within a quantile bin of log horsepower (Figure C.3a) and log number of workers (Figure C.3b). The line is a linear fit through the bin means.

D Appendix tables

Table D.1: Summary statistics by treatment and control group

		Share	Mean			
Variable	N	TG	TG	CG	Diff	t-stat
Baseline characteristics (1896, unless indicated otherwise)						
Extra-AHM trade	3,614	0.15	0.360	0.117	-0.243	-11.23***
Extra-regional trade (1898)	3,614	0.15	0.572	0.199	-0.373	-16.50***
All information available (worker, horsepower, output)	3,614	0.15	0.300	0.102	-0.199	-9.64***
Worker information available	3,614	0.15	0.625	0.279	-0.346	-15.37***
Number of workers	1,193	0.28	364.650	97.962	-266.688	-3.87***
Share of skilled workers	992	0.28	0.700	0.779	0.078	3.19***
Horsepower (HP) information available	3,614	0.15	0.447	0.134	-0.312	-13.93***
Installed horsepower	652	0.37	393.649	127.876	-265.774	-2.79***
Automation (HP/worker)	640	0.37	0.969	0.884	-0.084	-0.68
Output information available	3,614	0.15	0.317	0.121	-0.196	-9.34***
Output (in HUF)	541	0.31	1.43 mil	350,000	-1.08 mil	-5.13***
Labor productivity (output/worker, in HUF)	506	0.33	6,073	6,935	861.828	0.37
Multi-location firm	3,614	0.15	0.129	0.055	-0.075	-4.96***
Joint-stock company	3,614	0.15	0.141	0.021	-0.119	-7.80***
Firm age	1,391	0.24	27.355	25.519	-1.836	-1.04
Patent holder (1895)	3,614	0.15	0.099	0.084	0.15	-1.08
Patenting indicator (1896 - 1899)	3,614	0.15	0.144	0.103	0.042	-2.59***
Post-treatment outcomes (1904, unless indicated otherwise)						
Extra-regional trade	3,614	0.15	0.585	0.183	-0.402	-17.89***
Number of workers	796	0.36	372.180	89.312	-282.868	-4.04***
Extra-AHM trade (1905)	3,614	0.15	0.321	0.070	-0.251	-12.08***
Patenting indicator (1900 - 1905)	3,614	0.15	0.193	0.109	0.084	-4.67***

Table D.2: Number of exhibitors – Millennial Exhibition, Budapest 1896

Group No.	Name	No. of Exhibitors	Floor Area (m^2) Official	Private	Total	Outdoor Area (m^2)
a) Historical Exhibition		880	7,500	—	7,500	300
b) Contemporary Exhibition						
I/a	Fine Arts	269	3,600	—	3,600	—
I/b	Performing Arts	65	980	—	980	—
II.	Public Education & Culture	105	600	1,880	2,480	—
III.	Education	697	2,580	160	2,740	—
IV.	Public Health & Child Welfare	287	1,780	810	2,590	—
V.	Trade, Finance & Credit[1]	344	—	1,780	1,780	—
VI.	Agriculture	3,175	8,390	1,310	9,700	2,600
VII.	Forestry	214	1,500	90	1,590	7,500
VIII/a	Mining & Metallurgy	72	3,350	1,750	5,100	50
VIII/b	Iron & Metal Industry	231	1,760	90	1,850	300
IX.	Machinery & Engineering	269	13,380	4,050	17,430	500
X.	Transport, Navigation & Maritime	175	5,600	730	6,330	2,400
XI.	Construction Industry	286	2,200	1,340	3,540	500
XII.	Wood, Furniture & Decorative Arts	285	4,400	330	4,730	—
XIII.	Ceramics & Glass Industry	108	1,140	—	1,140	75
XIV.	Leather, Spinning, Weaving & Clothing	997	5,970	—	5,970	—
XV.	Paper & Printing Industry	191	—	2,250	2,250	—
XVI.	Gold, Silver & Decorative Goods	83	1,150	—	1,150	—
XVII.	Military Affairs[2]	44	3,110	450	3,560	2,500
XVIII.	Chemical Industry	114	950	860	1,810	—
XIX.	Food Products as Industrial Goods	323	990	3,180	4,170	—
XX.	Cottage Industry, Ethnography & Amateur Exh.	436	4,700	—	4,700	—
Total		**9,650**	**75,630**	**21,060**	**96,690**	**16,725**

[1] Organized ex officio.

[2] Joint Army, Navy, Home Guard, Gendarmerie, Red Cross. Organized ex officio.

[3] The floor area of village houses is included in this figure.

Table D.3: Pooled bounds regression: test of difference between upper bound, within, and lower bound

	(1) Extra-AHM trade 1905	(2) Extra-regional trade 1904	(3) ln(workers) 1904
Panel A: Upper-bound + within sample			
Paris exhibitor	0.101***	0.133***	0.230***
	(0.022)	(0.023)	(0.064)
Paris exhibitor × UB sample	0.082***	0.073***	0.011
	(0.017)	(0.012)	(0.045)
Observations	7,112	7,112	1,245
R-squared	0.474	0.312	0.192
Panel B: Lower-bound + within sample			
Paris exhibitor	0.042*	0.076***	0.163**
	(0.024)	(0.025)	(0.077)
Paris exhibitor × within sample	0.059***	0.057***	0.067
	(0.015)	(0.014)	(0.056)
Observations	6,264	6,264	1,182
R-squared	0.475	0.312	0.182
Baseline control (incl. × sample dummy)	Yes	Yes	Yes
Product FE (incl. × sample dummy)	Yes	Yes	Yes
UB and LB dummies	Yes	Yes	Yes

Robust standard errors in parentheses and clustered by product. Baseline control is extra-regional trade status 1898 in column (1), extra-AHM trade status 1896 in column (2), and ln workers 1896 in column (3); each control also interacted with the UB sample dummy. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To test whether the differences between adjacent bounds are statistically significant, we stack the relevant estimation samples pairwise and estimate a single interacted specification for each pair. We run two such tests: the upper bound against the within estimate, and the within against the lower bound estimate.

Consider a pair of stacked samples and let $S_i \in \{0, 1\}$ indicate that firm i belongs to the second sample of the pair, so that $S_i = 0$ denotes the reference sample. We estimate:

$$\begin{aligned} \Delta Y_i = \beta\, \mathrm{D}_i + \gamma\, (\mathrm{D}_i \times S_i) + \delta\, S_i \\ + Y_{i,pre} + Y_{i,pre} \times S_i + FE_p + FE_p \times S_i \\ + FE_c^0 \times (1 - S_i) + FE_c^1 \times S_i + \epsilon_i \end{aligned} \tag{D.1}$$

where D_i is the Paris exhibitor indicator, $Y_{i,pre}$ the baseline outcome and FE_p product category fixed effects, each included both in levels and interacted with S_i so that the covariate structure may differ across the two stacked samples. FE_c^0 and FE_c^1 denote the stratum fixed effects specific to the reference and second sample respectively, each active only within its corresponding subsample and set to zero elsewhere; the sample main effect δ is consequently absorbed.

In this specification β identifies the treatment effect in the reference sample, $\beta+\gamma$ recovers the effect in the second sample via a linear combination test, and a statistically significant

$\hat{\gamma}$ indicates that the two estimates are distinct. The two tests instantiate equation (D.1) as follows:

- **Upper bound versus within.** $S_i = U_i$ indicates membership in the upper bound sample and the within sample serves as the reference, so that FE_c^0 are jury award category fixed effects and FE_c^1 are the reverse comparison group fixed effects. Here β is the within estimate and $\beta + \gamma$ the upper bound estimate.

- **Within versus lower bound.** $S_i = W_i$ indicates membership in the within sample and the lower bound sample serves as the reference, so that FE_c^0 are the narrowly defined comparison group fixed effects and FE_c^1 are jury award category fixed effects. Here β is the lower bound estimate and $\beta + \gamma$ the within estimate.

Each stacked sample combines the two corresponding estimation samples as defined in equation (D.1), where the lower bound sample retains firms compared within narrowly defined comparison groups, the within sample retains firms compared within jury award categories, and the upper bound sample retains firms compared within reversed comparison groups on the full sample. Standard errors are clustered by product category.

Table D.4: The effect of the Paris exhibition on exporting: common sample

	Extra-AHM trade, 1905			Extra-regional trade, 1904		
	(1)	(2)	(3)	(4)	(5)	(6)
	Lower bound	Within	Upper bound	Lower bound	Within	Upper bound
Paris exhibitor	0.021	0.113***	0.178***	0.057*	0.137***	0.226***
	(0.025)	(0.024)	(0.034)	(0.032)	(0.027)	(0.033)
Lagged outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Common	Common	Common	Common	Common	Common
Observations	2,534	2,534	2,534	2,534	2,534	2,534
R-squared	0.489	0.489	0.486	0.317	0.317	0.317
Difference from within estimate	0.093***		0.065***	0.080***		0.089***
	(0.018)		(0.015)	(0.019)		(0.021)

Notes: This table repeats Table 4 on the common sample of treated firms in award classes 2 to 4 and the control firms each comparison assigns to them. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Table D.5: Treatment effects on patenting, by prior patenting status

	Patenting, 1900–1905		Patenting, 1896–1899	
	Upper bound	Lower bound	Upper bound	Lower bound
Paris × No prior patent	0.060***	0.040*	0.005	−0.016
	(0.016)	(0.023)	(0.017)	(0.024)
Paris × Prior patent	0.071	0.055	0.056	0.031
	(0.076)	(0.080)	(0.087)	(0.086)
Patenting status 1895	Yes	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes	Yes
Bound group dummies	Upper	Lower	Upper	Lower
Observations	3,614	2,650	3,614	2,650
R-squared	0.209	0.216	0.206	0.214

Notes: The outcome is an indicator variable that equals 1 if the firm applied for or registered at least one patent in the respective time period. Paris × No prior patent and Paris × Prior patent are interactions of the Paris exhibitor indicator with dummies for whether the firm had no prior patent or at least one patent registered by 1895, respectively. Bound group dummies refer to upper and lower bound sample restrictions. Standard errors are robust and reported in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

Table D.6 repeats Table 6 with one change: the buyer count adds French exhibitors, so that *foreign buyers* means all potential buyers in the firm's Paris class other than Hungarian ones. The main table counts only non-French, non-Hungarian buyers. The competition measures are unchanged, and *international competition* therefore continues to exclude both Hungarian and French competitors. French buyers are 22% of all buyers in the average Paris class.

Table D.6: Effect of competition and potential buyers on change in extra-AHM trade status, 1896–1905: French buyers included in the buyer count

	Extra-AHM trade, 1905					
	(1)	(2)	(3)	(4)	(5)	(6)
	Lower	Within	Upper	Lower	Within	Upper
Paris exhibitor	−0.061	0.026	0.033	0.028	0.076	0.188
	(0.099)	(0.081)	(0.095)	(0.112)	(0.101)	(0.124)
Paris exhibitor × ln(foreign buyer potential)	0.034***	0.033***	0.038***	0.036***	0.034***	0.041***
	(0.009)	(0.009)	(0.011)	(0.010)	(0.009)	(0.011)
Paris exhibitor × ln(Hungarian competition)	−0.047*	−0.056***	−0.048**	−0.031	−0.048**	−0.026
	(0.027)	(0.021)	(0.021)	(0.032)	(0.023)	(0.026)
Paris exhibitor × ln(international competition)				−0.024	−0.013	−0.038*
				(0.022)	(0.018)	(0.022)
Initial outcome	Yes	Yes	Yes	Yes	Yes	Yes
Product effects	Yes	Yes	Yes	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper	Lower	Within	Upper
Sample	Lower	Within	Upper	Lower	Within	Upper
Observations	2,650	3,614	3,498	2,650	3,614	3,498
R-squared	0.484	0.479	0.482	0.484	0.479	0.482

Notes: International buyers include French ones. Within compares treated and untreated firms within Budapest award classes; Lower and Upper use the adjacent-class comparisons of Section 4, so the estimation sample differs across columns. All specifications control for the initial outcome and include product fixed effects. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Each column estimates equation (4) and reports the implied buyer gradients: the change in the participation effect per additional log point of potential buyers in the firm's Paris class, separately for firms with and without a Paris award, and their difference θ. Panel A is the specification used in the main text. Panel B adds French exhibitors to the buyer count, so that buyers means all potential buyers other than Hungarian ones; French buyers are 22% of the buyer pool in the average class. Panel C drops the three firms that did not win a medal because they were not in competition.

Table D.7: The award premium and buyer thickness

	Extra-AHM trade, 1905		
	(1) Lower	(2) Within	(3) Upper
Panel A: Baseline: international buyers, hors concours *firms counted as having no award*			
Buyer gradient, awarded firms	0.049***	0.043***	0.050***
	(0.011)	(0.009)	(0.010)
Buyer gradient, firms without award	0.007	0.014	0.016
	(0.010)	(0.011)	(0.016)
Difference (θ)	0.042***	0.029**	0.035*
	(0.015)	(0.014)	(0.018)
Observations	2,650	3,614	3,498
Panel B: French buyers included in the buyer count			
Buyer gradient, awarded firms	0.049***	0.042***	0.049***
	(0.011)	(0.009)	(0.010)
Buyer gradient, firms without award	0.007	0.014	0.014
	(0.011)	(0.012)	(0.016)
Difference (θ)	0.042***	0.029*	0.035*
	(0.015)	(0.015)	(0.018)
Observations	2,650	3,614	3,498
Panel C: Hors concours *firms dropped from the sample*			
Buyer gradient, awarded firms	0.049***	0.043***	0.050***
	(0.011)	(0.009)	(0.010)
Buyer gradient, firms without award	0.012	0.018	0.022
	(0.012)	(0.012)	(0.017)
Difference (θ)	0.036**	0.025	0.029
	(0.016)	(0.015)	(0.019)
Observations	2,647	3,611	3,495
Baseline export status	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper

Notes: Within compares treated and untreated firms within Budapest award classes; Lower and Upper use the adjacent-class comparisons of Section 4, so the estimation sample differs across columns. All specifications control for baseline extra-AHM trade status and include product fixed effects. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Table D.8: Joint Treatment Effects on Exporting and Patenting (Lower Bound Estimates)

	(1) Extra-AHM trade	(2) Patenting, 1900-1905	(3) Both E.-AHM. & pat.	(4) Only E.-AHM.	(5) Only pat.	(6) Neither	(7) Either	Obs.
Panel A: Paris exhibitor × previous exporter								
Lower	0.113*	0.059*	0.070	0.043	−0.012	−0.102*	0.102*	2,650
	(0.065)	(0.034)	(0.044)	(0.052)	(0.039)	(0.051)	(0.051)	
Within	0.159***	0.056*	0.080**	0.078*	−0.024	−0.135***	0.135***	3,614
	(0.054)	(0.029)	(0.033)	(0.043)	(0.031)	(0.041)	(0.041)	
Upper	0.281***	0.087**	0.126***	0.156***	−0.039	−0.243***	0.243***	3,498
	(0.052)	(0.037)	(0.030)	(0.049)	(0.032)	(0.040)	(0.040)	
Panel B: Paris exhibitor × non-exporter								
Lower	0.024	0.037	0.011	0.013	0.026	−0.050	0.050	2,650
	(0.025)	(0.025)	(0.012)	(0.023)	(0.022)	(0.031)	(0.031)	
Within	0.083***	0.060***	0.036***	0.047**	0.024	−0.107***	0.107***	3,614
	(0.027)	(0.020)	(0.011)	(0.023)	(0.017)	(0.028)	(0.028)	
Upper	0.148***	0.087***	0.059***	0.089***	0.028	−0.176***	0.176***	3,498
	(0.036)	(0.021)	(0.015)	(0.025)	(0.018)	(0.034)	(0.034)	
Product fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Baseline export and patent status	Yes	Yes	Yes	Yes	Yes	Yes	Yes	

Notes: The outcome in column (1) is the change in extra-AHM trade status between 1896 and 1905; in column (2) it is an indicator for patenting in 1900–1905. Columns (3)–(7) report the probability of the indicated joint outcome, defined on extra-AHM trade status in 1905 and the patenting indicator: 'Both exp. & pat.' equals one for firms exporting in 1905 that also patented; 'Only exp.' for firms exporting in 1905 that did not patent; 'Only pat.' for firms not exporting in 1905 that patented; 'Neither' for firms doing neither; and 'Either' is one minus 'Neither'. The four categories are mutually exclusive and exhaust the sample. Because all specifications control for baseline extra-AHM trade status, the treatment coefficient on the 1905 level equals that on the change, so column (1) and the sum of columns (3) and (4) coincide by construction. All specifications also control for baseline patenting status and its interaction with baseline extra-AHM trade status, and include product fixed effects. Bound group dummies refer to lower bound sample restrictions. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Table D.9: The effect of the Paris exhibition on employment growth: common sample

	ln(workers), 1904		
	(1) Lower bound	(2) Within	(3) Upper bound
Paris exhibitor	0.100	0.176**	0.258***
	(0.108)	(0.072)	(0.072)
Pre-treatment outcome	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper
Sample	Common	Common	Common
Observations	495	495	495
R-squared	0.173	0.173	0.172
Difference from within estimate	0.077		0.082**
	(0.063)		(0.042)

Notes: This table repeats Table 10 on the common sample of treated firms in award classes 2 to 4 and the control firms each comparison assigns to them. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

Table D.10: The effect of the Paris exhibition on extra-regional trade in 1909: common sample

	Extra-regional trade, 1909		
	(1) Lower bound	(2) Within	(3) Upper bound
Paris exhibitor	0.031	0.115***	0.204***
	(0.036)	(0.026)	(0.031)
Pre-treatment outcome	Yes	Yes	Yes
Product fixed effects	Yes	Yes	Yes
Bound group dummies	Lower	Within	Upper
Sample	Common	Common	Common
Observations	2,534	2,534	2,534
R-squared	0.476	0.476	0.476

Notes: This table repeats Table 12 on the common sample of treated firms in award classes 2 to 4 and the control firms each comparison assigns to them. Standard errors are clustered by product. *** p<0.01, ** p<0.05, * p<0.1.

E Exhibition Rules and Institutional Framework

E.1 Geographic composition of the Austrian Hungarian Monarchy

The Austrian Hungarian Monarchy composed of different parts, as illustrated in Figure E.1.



Figure E.1: GDP per capita, 1870–1913

Source: Authors' elaboration based on A. Piechl and P. P. Marckhgott-Sanabria, "Austro-Hungarian Empire Crownlands 1910," HistoGIS, ACDH-CH, Austrian Academy of Sciences, https://histogis.acdh.oeaw.ac.at/shapes/source/detail/141 (accessed 21 July 2026), CC BY 4.0. Boundaries generalized; Syrmia merged into Croatia-Slavonia and the Fiume feature corrected by the authors.

F Exhibition Rules and Institutional Framework

The credibility of our empirical strategy relies on understanding the institutional rules governing both exhibitions. The Budapest exhibition provided an objective measure of export potential through its independent jury system, while the Paris exhibition's selection constraints created the variation necessary for identification.

F.1 Budapest 1896 Millennial Exhibition Rules

The 1896 Budapest exhibition was designed as a comprehensive showcase of Hungarian industry and served as a "trial run" for identifying firms with export potential. The exhibition attracted over 9,000 exhibitors across all sectors, with mandatory evaluation of all commercial exhibits by an independent jury system.

F.1.1 Jury Independence and Structure

The jury operated independently from any future selection processes. According to the official regulations, jury members were selected through a mixed process: half elected by exhibitors within each industry group, half appointed by the exhibition committee in consultation with relevant ministries. The jury consisted of 480 members across 34 specialized industry groups, with each group electing its own president, vice-president, and rapporteur by secret ballot and simple majority.

Crucially, jury members and their associated firms were explicitly excluded from receiving awards in their respective groups, ensuring objective evaluation. Experts serving in advisory roles were similarly excluded from awards in the specific groups where they provided assistance. State authorities, institutions, and enterprises were also ineligible for awards, with their items marked with appropriate labels.

F.1.2 Award Categories and Hierarchy

The jury awarded four distinct categories of recognition, listed below in descending order of prestige. Our labels used in the main text are given in parentheses.

1. **Honorary Diploma (Gold)**: The highest award, given to exhibitors who had earned "national merit" through their contributions, awarded by the jury council based on group jury proposals. A maximum of 100 Honorary Diplomas could be granted. Since some exhibits were jointly produced by several firms, the 100 diplomas correspond to 135 unique firms in our sample.

2. **Millennial Grand Medal (Silver)**: Awarded to exhibitors who "demonstrated exceptional advancement in their field, elevated their industry, introduced new types of production or methods, contributed to mass production or exports, and enhanced the prestige of the exhibition through the excellence of their exhibits."

3. **Exhibition Medal (Bronze)**: Awarded to exhibitors who demonstrated merit worthy of distinction.

4. **Certificate of Recognition (Honorable Mention)**: Granted to exhibitors not eligible for higher awards but whose work deserved acknowledgment.

Except for honorary diplomas, there was no limit to the number of medals and certificates awarded. Each medal was accompanied by a certificate listing the specific merits upon which the award was based.

F.1.3 Award Criteria

The regulations often explicitly specified export capability as a primary evaluation criterion. Common citations in the jury's stated rationales for awards included: "good/excellent export capability" (kivitelképesség), "competitiveness" (versenyképesség), "introducing a new industry in Hungary (új iparág bevezetése)," "good/excellent work/effort (jó/kiváló munka), or performance (teljesítmény)," and "good/excellent progress or innovation (jó/kiváló haladás)." The emphasis on export potential was central to the exhibition's role as a trial run for identifying firms suitable for international representation.

F.1.4 Evaluation Process

All exhibited items underwent mandatory evaluation unless the exhibitor explicitly declined within a set deadline. The jury conducted systematic on-site inspections and decided for each exhibitor: (1) whether they should receive a distinction and why, (2) which award should be granted, and (3) what specific citation should be used.

Group jury decisions were made by majority vote, with the president's vote deciding in case of ties. When decisions were not unanimous, minority opinions could be recorded upon request. The evaluation process was conducted under strict confidentiality, with no appeals permitted against jury council decisions.

F.2 Paris 1900 Exhibition Rules and Selection Process

The Paris exhibition operated under a sophisticated three-tier international jury system, but Hungarian participation was determined by a separate domestic selection process subject to multiple constraints beyond export potential.

F.2.1 Scale and Commercial Purpose

The 1900 Paris Exposition Universelle was explicitly designed as the largest international commercial gathering to date, featuring 76,000 exhibitors from 56 countries across 121 product subcategories. The exhibition's commercial objectives were formally recognized, with contemporary documentation describing it as providing "favorable opportunities for establishing commercial connections" and facilitating face-to-face meetings between international buyers and sellers.

F.2.2 Hungarian Selection Constraints

While the primary goal was promoting exports by selecting high-potential firms, internal government records reveal three additional constraints that prevented perfect sorting based solely on export potential:

1. **Space Limitations**: Limited pavilion space in Paris restricted total participation numbers.

2. **Representational Balance**: The selection committee sought balanced representation across industries and regions to present a comprehensive picture of Hungarian production capabilities.

3. **Transport Costs**: High costs for shipping exhibits, particularly heavy machinery and bulky products, influenced selection decisions and created logistical constraints.

F.2.3 Product Category Organization

Exhibitors were classified into 18 broad categories and 121 subcategories based on their specific exhibits rather than their full product portfolios. This classification system determined competitive exposure at the exhibition, with subcategory sizes varying from hundreds to thousands of exhibitors. Categories ranged from narrowly defined products (e.g., "shoes") to broader industrial classifications (e.g., "chemicals").

F.2.4 Commercial Networking Framework

The exhibition operated from April to November 1900, providing extended opportunities for commercial relationship formation. The 48 million visitors included merchants, industrialists, and potential customers, creating an unprecedented concentration of international commercial contacts. Contemporary accounts emphasize active commercial negotiations occurring alongside product displays throughout the exhibition grounds.

These institutional details underscore that Budapest awards provide a credible measure of export potential determined by independent expert evaluation using explicit export-focused criteria, while Paris selection involved strategic balancing across multiple objectives, creating the selection patterns we exploit for identification.

G Data Appendix

This appendix documents the data collection, processing, and variable construction procedures for the analysis of Hungarian firms' exhibition participation.

In this appendix, we discuss the key steps awhile acknowledging inherent limitations of historical administrative sources.

G.1 The 1896 Budapest Exhibition

Budapest Exhibition 1896 catalog Complete official catalog of the 1896 Millennial National Exhibition containing 4,626 exhibitor records across contemporary industrial categories (agriculture excluded). Source: *Az 1896-iki Ezredéves Országos Kiállítás általános katalógusa*, edited by Mudrony Soma, Ráth Károly, and Micseh Endre (Budapest: Kosmos Ny., 1896), shelf number 282.068. The catalog comprises 21 category-specific volumes, each with independent numbering systems.

Jury Awards 1896. Official jury evaluations from the Budapest exhibition, documented in *Az 1896-iki ezredéves országos kiállítás közleményei*, issue 32, pages 1229-1236 (shelf number H 2.008 FM3/4736). Awards categorized hierarchically: Grand Certificate, Grand Medal, Exhibition Medal, and Certificate of Acknowledgment.

G.1.1 Filtering

To focus on commercially oriented manufacturing firms, we apply three filters to the full catalog. First, we retain only contemporary exhibition groups within manufacturing, excluding groups such as Group I/a (Fine Arts), III (Education), or V (Trade, Finance, and Credit), where no for profit manufacturing firms are present. Second, within these groups, we exclude non-commercial entities (municipalities, associations, and NGOs) and consolidate duplicate entries where firms exhibited across multiple categories. Third, we restrict the sample to firms that entered the jury competition for awards, which excludes jury members (70 firms) and exhibitors that explicitly asked to be excluded from competition (72 firms).

After applying these filters, the sample comprises 3,736 for-profit manufacturing firms. Not all of these firms survive until the end of our sample period in 1904–1905. To estimate treatment effects conditional on survival, we identify firm exits between 1896 and 1904 by matching firm names to the exit reports from the *Central Gazette of Hungarian Firms*.

G.2 The 1900 Paris Exhibition and Trade Fair

Paris Exhibition 1900 catalog Hungarian-language catalog of Magyar exhibitors at the 1900 Paris International Exhibition, documenting 2,665 participants with detailed category and subcategory classifications. Source: *Az 1900. Párisi Magyar Kiállítás kalauza: Magyarország ismertetése* (Budapest: Riegler Ny., 1900), shelf number 262.540. This catalog serves as the primary treatment indicator for the empirical analysis.

G.3 Data Digitization and Text Processing

G.3.1 Digitization Protocol

All primary sources accessed in printed format underwent systematic digitization: (1) photographic capture using Hungarian National Library equipment, (2) optical character recognition via Google Document AI, and (3) systematic error correction through comparison with original printed books.

G.3.2 Text Processing Pipeline

Raw OCR output processed using Python with regular expression libraries implementing:

1. **Character encoding standardization:** Conversion of Hungarian diacritical marks and historical orthographic conventions to Unicode UTF-8 format
2. **Field extraction:** Parsing of semi-structured text entries into discrete variables using pattern recognition algorithms tailored to each source's formatting conventions
3. **Manual verification:** Systematic correction of automated extraction errors, with particular attention to firm names, addresses, and numerical data

G.4 Exhibition Participation and Awards

G.4.1 Exhibition Variables

Table G.1: Exhibition Participation Variables

Variable	Definition and Construction
`exhibition_category_YYYY`	Industry classification within exhibition framework. Original categorization preserved with harmonized coding for cross-temporal analysis.
`exhibition_subcategory_YYYY`	Detailed product-level classification within broader industry categories. Enables fine-grained analysis of exhibition participation patterns.
`exhibited_product_YYYY`	Specific products displayed at exhibition, distinct from firm's full product portfolio. Captures strategic decisions regarding exhibition representation.

G.5 Entity Resolution and Panel Construction

G.5.1 Matching Methodology

Panel dataset construction required systematic entity resolution across five independently collected datasets. The matching procedure implemented:

Pairwise Matching Algorithm:

1. **String similarity:** Jaro-Winkler distance calculation for firm names with threshold optimization balancing precision and recall
2. **Geographic consistency:** Exact and approximate location matching using standardized municipal codes
3. **Product similarity:** Jaccard similarity coefficients for product portfolio overlap
4. **Temporal consistency:** Validation of firm characteristic evolution (employment, technology) for plausibility

Correspondence Tables: Five pairwise matching tables documenting confirmed firm linkages:

- 1896 Budapest Catalog ↔ 1898 Export Registry
- 1896 Budapest Catalog ↔ 1900 Paris Catalog
- 1896 Budapest Catalog ↔ 1904 Export Registry
- 1898 Export Registry ↔ 1904 Export Registry

Master Identification Table: Comprehensive mapping consolidating direct and indirect matches to enable multi-period panel construction with both balanced and unbalanced specifications.

G.5.2 Match Quality and Coverage

Table G.2: Entity Resolution Results

Dataset Pair	Total Potential Matches	Confirmed Matches	Match Rate (%)
1896 Budapest ↔ 1898 Export	1,894	1,059	55.9
1896 Budapest ↔ 1900 Paris	2,665	665	24.9
1896 Budapest ↔ 1904 Export	4,381	957	21.8
1898 Export ↔ 1904 Export	1,894	1,114	58.8
1896 Budapest ↔ 1909 Export	3,941	547	13.8

G.6 Sample Characteristics and Descriptive Statistics

G.6.1 Dataset Coverage and Temporal Scope

Table G.3: Dataset Characteristics

Dataset	Observations	Time Period	Processing Status	Key Innovation
Budapest 1896	4,626	1896	Complete*	Universal exhibition census
Export 1898	1,894	1897-1898	Complete	Export-ready classification
Paris 1900	2,665	1900	Complete	International treatment
Export 1904	4,381	1904-1906	Complete	Comprehensive registry
Export 1909	3,941	1909	Complete	Comprehensive registry

*Excludes education and agriculture categories

G.7 Measurement Limitations and Econometric Considerations

G.7.1 Historical Source Constraints

Measurement Error: Firm-reported employment and capital measures subject to strategic reporting incentives and standardization challenges across heterogeneous enterprises.

Survival Bias: Panel analysis necessarily conditions on firm survival through the observation period, potentially understating treatment effects for marginal firms.

G.7.2 Entity Resolution Limitations

Several issues arose during the process that may generate both types of errors. We looked at matches manually but it remains possible.

Matching Precision: Name variations, mergers, and reorganizations

Geographic Mobility: Firm relocation between observations may complicate identification. (A systematic address tracking suggests limited geographic mobility for manufacturing enterprises.)

G.7.3 Variable Construction Issues

Industry Classification Harmonization: Evolving industrial classification systems across datasets necessitate concordance tables with potential misclassification of diversified firms.

Temporal Aggregation: Registry publication dates may not correspond precisely to data collection periods, introducing timing measurement error requiring robustness checks with alternative temporal specifications.

G.8 Patents

G.8.1 Datasets

We combine patent data from several complementary historical sources:

- **Austrian imperial privileges.** Before the establishment of the Austrian and Hungarian patent offices, firms were able to obtain 'imperial privileges'. We obtained Austrian privilege records covering 1837 to 1899 from the Privilegiensammlung des österreichischen Patentamts from the Austrian National Library.
- **Hungarian Patent Office.** The Hungarian Patent Office was established in 1896. We retrieved historical patent records covering 1887 to 1920 from the Hungarian Patent Office online database via web scraping.
- **PATSTAT.** The Austrian National Library provided us with historical Austrian patent applications covering 1890 to 1925.
- **Google Patents.** We complement the Austrian and Hungarian sources with Google Patents for Austria and several other large countries (to account for the possibility of Hungarian firms registering patents abroad): United States, Germany, France, Great Britain, and Switzerland, covering all data up to 1925 to capture long-term outcomes.
- **Google Patents (long-term outcomes, Hungary).** To analyze long-term outcomes, we additionally downloaded Hungarian patent records from Google Patents covering 1980 to 2014.

All patent sources are first harmonized into a common long-format dataset. Each record receives a patent identifier, source label, matched name, source country or source group, and an event year. The event year is the year used to assign a patent record to time. For Google Patents, it is based on the publication date. For Austrian PATSTAT filings, it is based on the filing date, falling back to the filing year and, where neither is available, to the earliest publication year. For Austrian imperial privileges, it is based on the grant date. For Hungarian Patent Office records, it is based on the application date where available, otherwise on the publication date.

Only observations with a valid event year between 1800 and 1925 are used for variable construction; the long-term Google Patents data for Hungary (1980–2014) is used separately for the long-term outcomes analysis described below. After matching patent records to firms in our sample, patent duplicates with the same source-specific patent identifier are removed within each firm. This means that repeated matches caused by multiple names, applicants, inventors, or alias variants are counted only once for the same firm. The main firm-level variables are constructed based on the remaining matched records.

G.8.2 Matching firms to patents

Matching firms to patents is implemented as a two-stage pipeline. The first script constructs recall-oriented candidate matches between Budapest firms and patent records. The second script applies stricter validation rules and rebuilds the final firm-level panel.

Names are normalized by removing accents, punctuation, capitalization differences, generic legal and business terms, and selected historical spelling variants. The algorithm constructs multiple name representations, including canonical names, core names, score names, head tokens, strong tokens, and anchor tokens. Candidate retrieval uses indexes over these representations and, for firms with independent product evidence, patent-title tokens. Matching is run first without aliases on the firm side and then, for previously unmatched firms, with English, German, and French alias variants of common firm-name components such as company, factory, brothers, sons, widow, royal, and joint-stock company; the same alias translation is applied when building the patent-side data. Candidate scores combine exact-name agreement, fuzzy string similarity, token overlap, weighted anchor-token overlap, head-token agreement, and product-title context.

The verification stage re-scores candidates and classifies them. Accepted matches require exact full or core-name agreement, a single-surname match supported by a matching given name, high fuzzy similarity with sufficient core-token overlap, rare-token evidence with initials or product context, or known-positive-specific evidence. Candidates are rejected for weak name evidence, generic-token matches, unsupported single-surname matches, or given-name conflicts.

For the overlapping period from 1896 to 1905, results produced by different versions of the matching algorithm were additionally compared and manually reviewed; correct matches were retained and incorrect or inconsistent matches were discarded, regardless of which algorithm version had produced them.

Accepted matches are de-duplicated at the firm-patent-source-year level. This removes repeated matches caused by aliases, multiple applicants or inventors, and alternative name variants. It does not fully de-duplicate the same underlying invention across different sources if different identifiers are used.

G.9 Competition and Buyer Potential

G.9.1 Raw Data

The raw input data used to estimate competition and market thickness originates from the official catalogs of the product groups, comprising 20 volumes[38]. The exposition is organized into eighteen groups; some groups are split across multiple volumes because of the very large number of exhibitors in particular subcategories (for example, the wine producers). For each volume we use the website's OCR option to download a raw text file. Each book contains multiple subgroups, called *classes*, which correspond approximately to product categories (121 in total). Within each class, exhibitors are arranged by country: French exhibitors come first (usually the largest share), followed by the French colonies, then the remaining countries in alphabetical order. For every exhibitor the entry lists the firm or individual's name and the exhibited products.

The total number of listings across the 121 classes is 114,674 exhibitors, including firms, artisans, public institutions, and municipalities, but excluding artists and historical exhibits. About 30% of this are French firms including its colonies. The remaining exhibitors come from all around the world, however, due to the location of the exhibition, European countries are overrepresented.

G.9.2 Deriving the Paris product category-Level Summaries from Text Data

Even before assessing which Paris product category may be a potential buyer of supplier of another Paris product category, we need to understand the type of firms exhibiting in each. We use LLMs to extract relevant part of the text for each Paris product category and to create a structured description of its jury overview and a list of exhibitors types (e.g. "Automobiles (petrol-engined)" or "Luxury horse-drawn carriages" in Paris product category 30.). The jury overview is summarized in a short text and contains information on the history of the products exhibited in the Paris product category, current technological trends, and exhibition specific information. The list of exhibitor types contains besides the firm type (e.g. "Automobiles (petrol-engined") a short description of examples (e.g. "omnibus and family wagon automobiles; hunting and sporting cars; heavy-load road trains"). The Paris product category-level summarized, containing the lists of exhibitor types and jury text is the immediate input for the input-output analysis used to estimate the number of potential buyers for each Paris product category.

Two implications follow for the way these summaries are used in the construction of the buyer scores. First, the files are *secondary* sources: they encode a single model's structured reading of the OCR'd jury texts, not the jury texts themselves. Any Paris product category-level claim that materially affects a matrix cell — in particular evaluative phrasing such as "the single largest category" or quantitative shares implicit in the jury overview — is checked against the underlying OCR before being treated as authoritative. Second, the formatting is regular enough that the Paris product category headers and the bolded category bullets can be parsed deterministically into *(Paris product category number, category name, exemplar firms)* triples, which would allow the calculation of inputs demanded by the firms in the

[38]Scanned versions of the original catalogs are available at Internet Archive

subcategory. This is the route taken in the backward-link audit, where the bullet categories were used as the unit of evidence for whether a candidate supply link *exists* (P_{exists}).

G.10 Measure of Competition

To estimate the intensity of competition firms faced when exhibiting in Paris, we first count the number of participants in each of the Paris product category. This number is a good indicator of competition in homogeneous examples (e.g. wines) but not in heterogeneous ones. For example, in Paris product category 11, a firm exhibiting a type of foundry does not compete with other firms showing printers; or in Paris product category 65, a brass-fittings maker is an input supplier to plumbers in its own Paris product category rather than their rival. We, therefore, decompose each firm's co-exhibitors in the same Paris product category into a competitor pool and a potential-buyer pool using an LLM. For each firm i we segment other exhibitors into two groups, namely an own-subcategory buyer share $\pi_j \in [0, 1]$, defined as the share of other firms in the same product subcategory whose product profile makes them plausible buyers of j's output rather than direct competitors, and a competitor share $1-\pi_j$. The firm-specific buyer share is assigned based on the firm's product description and the information about which products are used as inputs in other products described below in Section G.11.1, as well as the composition of different types of firms documented in the product category summary (e.g., component makers, integrators, end-product sellers).

The scale of π_j is discrete, between 0.10 and 0.88, and denotes graded confidence, not a calibrated probability — so the ranking of companies is more reliable than the levels themselves. Two limitations are worth noting. First, the correction only adds anything in vertically segmented Paris product categories: in homogeneous ones (wine, footwear, basket-weaving) almost every company sits at the lower bound of 0.10, where the variable effectively collapses back to bare Paris product category size, and the meaningful variation comes from Paris product categories of the typography, light-metal, chemical-industry type. Second, the estimate looks at the company's own product description, not at the other exhibitors', so within-Paris product category variation arising from exactly what the non-Hungarian exhibitors produced is not captured. 533 companies received a value across 71 Paris product categories, mean 0.19, median 0.10.

The competition variable used in regressions is constructed using the buyer share π_j and the number of other firms in each firms' own product subcategory, decomposed as follows: N_Hun_j for Hungarian firms and N_Int_j for international (non-local) firms. Thus,

$$\text{Hungarian Competition}_j = (1 - \pi_j) \times N_Hun_j$$

International competition is calculated the same way.

G.11 Measure of Buyers

G.11.1 Estimating buyer-supplier relations between Paris classes

Using the groups summaries described in Section G.9.2, we use an LLM to create a 121x121 table shoring probabilities of input-output linkages. Each cell (j, k) records a directed supply relationship running from a supplying Paris product category j to a buying Paris product

category k. Because no monetary transaction data survives for this population, the cell entry is a probability, P_{jk}: the probability that among all firms exhibiting at Paris, at least one pair existed in which a firm in Paris product category j supplied inputs to a firm in Paris product category k. The quantity is explicitly conditioned on the exhibitor population rather than on the 1900 world economy—a value of 0.90 states confidence that the relationship was present among Paris exhibitors, not that 90 per cent of firms worldwide had such a link—so Paris product categories densely represented at the exposition accumulate more documented relationships than niche or peripheral ones. An empty cell therefore records only that no linkage was identified, not that none existed.

The evidence base is the OCR text of the twenty-volume *Catalogue Général Officiel*, introduced in Section G.9.2. Working Paris product category by category, the language model read this material and produced a structured summary of each Paris product category's position in the production network: whether the Paris product category was a raw-material supplier, an intermediate processor or an end-product manufacturer; the inputs it bought (backward linkages); the outputs it sold and to whom (forward linkages); firms exhibiting in more than one Paris product category, taken as evidence of integration across the supply chain; and a full enumeration of exhibited products. The reasoning that established a linkage ran from the exhibited product back to the material it must be made from, and then to the Paris product category that produces that material. Where the catalogue records Société Parvillée showing '*Porcelaines et ferrures pour l'électricité. Matériel d'isolateurs*' in the electrochemistry Paris product category, the porcelain insulators entail a purchase from the ceramics Paris product category; Saint-Gobain's '*Bacs d'accumulateurs. Moulages en verre*' entail a purchase from glassware; the copper founding, refining and rolling of the Compagnie française des Métaux entails a purchase from heavy metallurgy. The inference is thus anchored in what a named firm states that it makes, and a linkage is asserted between two Paris product category only where the product description or the jury narrative identifies the input concretely enough to name the supplying Paris product category.

Each linkage was tagged at the point of identification with an evidence type—direct catalogue quote, jury-report narrative, or derived from product logic—and with a confidence label of HIGH or POSSIBLE, and these two tags jointly determine the numeric probability. The ordering reflects the evidentiary distance between the text and the claim: a quote from an exhibitor's own entry is firm-level testimony and receives the highest value (0.90); a jury report is reliable but constitutes industry-level rather than firm-level documentation (0.80); and a relationship nowhere stated but following unambiguously from the products exhibited—a rubber-goods manufacturer must buy raw rubber—is confident yet inferred (0.65). The downgrade from HIGH to POSSIBLE applies where the direction of the relationship is uncertain, where its regularity is unclear, or where it may hold for some firms in a Paris product category but not for all, yielding 0.60, 0.55 and 0.45 for quote, jury and derived evidence respectively. Links were then harvested from the summaries mechanically rather than by further judgement: pattern matching captured every arrow expression of the form '→ Paris product category N (description), confidence: HIGH/POSSIBLE', reading the evidence type from the preceding text and the direction from whether the expression appeared in a backward or a forward linkage section. The resulting distribution sits heavily at the top of the ladder, with roughly two-thirds of all links at 0.90—a direct consequence of summaries built to prioritise quote-backed relationships.

We define cross-Paris product category *Buyer Potential* as a probability weighted sum of potential international buyers. In the main model, we excluded French firms since an increase in French imports from Hungary might result from learning about the French market outside the scope of the exhibition. We then use these probabilities to weight the number of potential buyers each Hungarian exhibitor faced. For a Hungarian firm exhibiting in Paris product category j, each Paris product category k to which j has a forward link contributes $P_{jk} \times N_k$ potential buyers, where N_k is the number of exhibitors in Paris product category k; summing over all such k gives $\sum_{k\,:\,j\to k} P_{jk} \times N_k$.

The version of *Buyer Potential* used in the regressions adds buyers in other Paris Paris product category to buyers inside the firm's own Paris product category. For a firm in Paris product category j, each Paris product category k to which j has a forward link contributes $P_{jk} \times N_k$ expected buyers. This is summarized as follows:

$$\text{Buyer Potential}_j = \sum_{k\,:\,j\to k} P_{jk} \times N_k + \pi_j \times N_j$$

As described above, we are mainly interested in international buyer potential (excluding French and Hungarian potential buyers) or, as an alternative, foreign buyer potential (only excluding Hungarian potential buyers).

G.12 Long-run survival

Six of the Paris exhibitors still operate today. *Zsolnay* of Pécs is a ceramics manufactory whose eosin lustre glaze and frost-resistant pyrogranite made it one of the most visible Hungarian producers in Paris, nationalised in 1948 as the Pécs Porcelain Factory and re-established as Zsolnay Porcelánmanufaktúra Zrt. under municipal ownership. Its founder, Vilmos Zsolnay, died in March 1900, weeks before the exhibition opened, and his son Miklós presided over the Paris exhibit. *Herend*, which appears in our 1896 catalogue under the name of its then proprietor Farkasházi Fischer, followed a similar path and operates today as Herendi Porcelánmanufaktúra Zrt. *Zwack*, the Budapest distiller of the Unicum bitter, was nationalized in 1948 and re-acquired by the family in 1991; it is now listed on the Budapest Stock Exchange. *Pick* of Szeged, the winter salami maker, was nationalized in 1948, privatized in 1992 and trades today as Pick Szeged Zrt. *Gerbeaud*, a Budapest confectioner, was nationalized after 1948 and traded as the Vörösmarty confectionery until its name was restored in 1984. The final firm, *Thonet*, is the most prominent: the Hungarian house in our catalogue was one arm of a multinational family firm whose main plants in historic Hungary lay in territory transferred to Czechoslovakia in 1920 and were nationalized in the 1940s. What survives today is the German branch together with a separately owned Viennese brand.

Two of the firms that did not exhibit in Paris still operate today. *Kner Izidor*'s printing house in Gyoma, founded in 1882, was nationalized in 1949 and operates today as Gyomai Kner Nyomda Zrt., part of a listed printing group. *Auguszt*, a confectioner founded in Buda in 1870, is still run by the founding family, its operation interrupted by nationalization between 1951 and 1957.